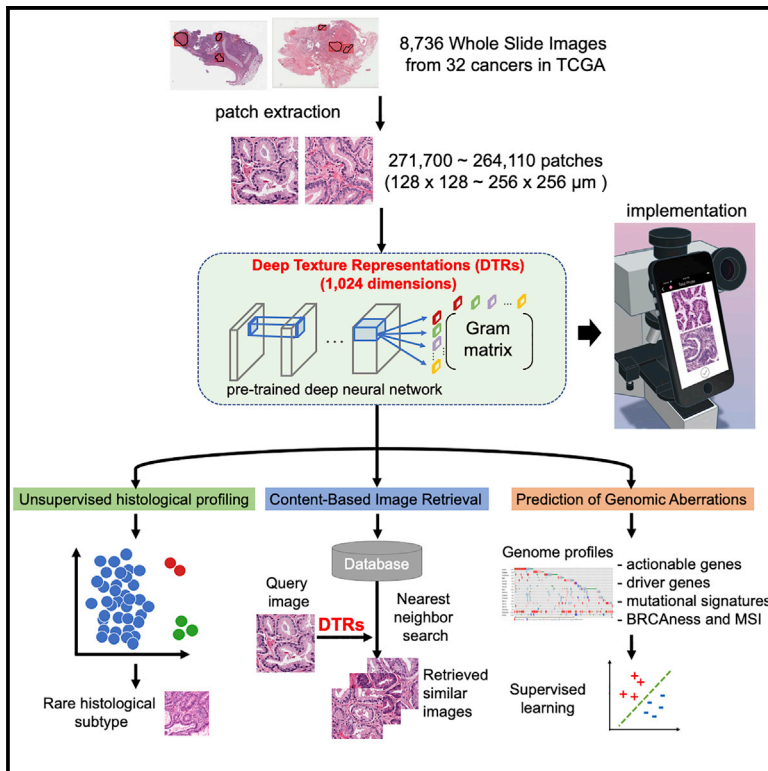


Universal encoding of pan-cancer histology by deep texture representations

Graphical abstract



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In brief

Komura et al. develop a method to encode cancer histology using a deep neural network. The utility of the method is validated in three clinically useful applications: unsupervised analysis of cancer morphology to detect histological subtypes, retrieving histologically similar images among large datasets, and predicting genomic features from histology images

Highlights

- DTR is a universal encoder for cancer histology using a deep neural network
- DTR enables quick and accurate retrieval of histologically similar images
- DTR enables genomic feature prediction from histology images in various cancer types
- DTR-based image retrieval system is publicly accessible from PC and smartphone



Resource

Universal encoding of pan-cancer histology by deep texture representations

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SUMMARY

Cancer histological images contain rich biological and clinical information, but quantitative representation can be problematic and has prevented the direct comparison and accumulation of large-scale datasets. Here, we show successful universal encoding of cancer histology by deep texture representations (DTRs) produced by a bilinear convolutional neural network. DTR-based, unsupervised histological profiling, which captures the morphological diversity, is applied to cancer biopsies and reveals relationships between histologic characteristics and the response to immune checkpoint inhibitors (ICIs). Content-based image retrieval based on DTRs enables the quick retrieval of histologically similar images using The Cancer Genome Atlas (TCGA) dataset. Furthermore, via comprehensive comparisons with driver and clinically actionable gene mutations, we successfully predict 309 combinations of genomic features and cancer types from hematoxylin-and-eosin-stained images. With its mounting capabilities on accessible devices, such as smartphones, universal encoding for cancer histology has a strong impact on global equalization for cancer diagnosis and therapies.

INTRODUCTION

For more than a century, pathologists have observed tissue slides via microscope not only to diagnose diseases but also to uncover clues for understanding disease pathophysiology. Careful observation has led to the discovery of numerous biologically and clinically relevant morphological features of both pathogens and host tissues. However, one critical limitation of this approach is that there have been few ways to objectively and quantitatively express histological features, making it difficult to communicate with others among research/clinical communities, and to compare the data directly with other cases. Although there are some methods to do that in some specific cancers (e.g., estrogen/progesterone receptor status in breast cancers using immunohistochemistry), it only captures one aspect of the histological features, and the method to universally quantify the complex morphology is required. This situation strongly contrasts with that of genomic data, for which researchers can directly compare as well as accu-

mulate and retrieve new findings from thousands of cases. As such, investigations like genome-wide association studies have a substantial impact on current biological and clinical sciences.

Application of the convolutional neural network (CNN), a deep-learning technology, has thus far been successful for the classification, detection, and segmentation of histopathology images. CNN has already surpassed human pathologists in terms of accuracy and diagnostic times within some applications (Bejnordi et al., 2017; Nagpal et al., 2018). However, even though this approach is powerful, CNNs require a task-specific feature extraction process for individual tasks, meaning that a huge amount of training data is essential to construct trained networks to cover every subject in histopathology fields. This setback hampers the widespread application of artificial intelligence in pathology communities. Another way to utilize the CNN within histopathology is through its use as a general feature extractor (Hatipoglu and Bilgin, 2016). However, unlike general images like those of cats and dogs, morphological features of histopathology images—especially for



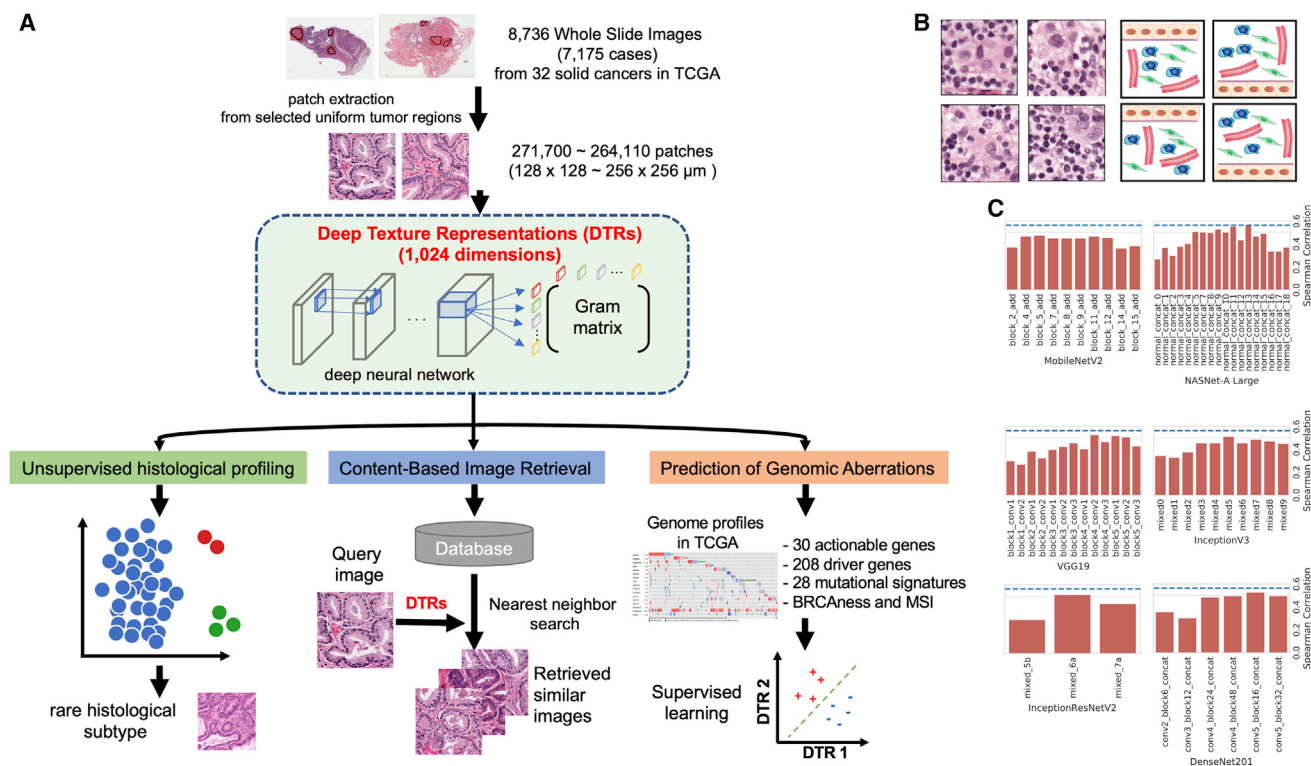


Figure 1. Universal encoding of pan-cancer histology by deep texture representations (DTRs)

(A) Workflow overview. Uniform tumor regions within 8,736 whole-slide images representing 32 solid cancers from TCGA dataset were annotated by trained pathologists, and 256 $\times$ 256 pixel patches (128 $\times$ 128 ~ 256 $\times$ 256 μm in actual size) were generated. By utilizing the optimal pretrained CNN, 1,024-dimensional vector representations were extracted for each patch and tested in three applications: unsupervised histology profiling to identify histological subtypes, CBIR to retrieve histologically similar images, and prediction of genomic aberration from histopathology images.

(B) Examples of histologically similar or dissimilar representative images (left) and pattern diagrams (right). Although each cell type's relative positions were different among the two images on the left, they should be regarded as similar (positional invariance). In contrast, the two images on the right have tissues with different cell compositions or different cell types, and thus they should have a different texture from those on the left. The pattern diagrams, including five various cell types (tumor cells, lymphocytes, fibroblasts, endothelium, and normal epithelium) depict the same things for clarity.

(C) Spearman's rank correlation between similarity of images ranked by pathologists and DTRs. The layers are deeper along the x axis from left to right in each network. DTRs from middle layers in NasNet-A Large network showed the strongest association to the similarity ranked by pathologists.

cancer histology—consist of spatial invariance, such as the texture(s) derived from cellular and extracellular components, rather than their relative positions within the image, as observed in typical photographs (Masaki, 2006). This phenomenon has prompted us to develop a feature extractor that can express tissue morphology objectively and quantitatively.

Here, we showed that deep texture representations (DTRs) produced by bilinear CNN (B-CNN) (Lin et al., 2015) trained on 1.2 million general images could adequately express cancer morphology information without extra training on histology images, and by selecting an optimal combination of a network model and its layer, this functions as a universal encoder for cancer histology (Figure 1). DTRs are based on a B-CNN, which enables the extraction of spatially invariant features independent of the relative location of individual cellular and extracellular components within the image (Figures 1A and 1B). We then used compact bilinear pooling (CBP) (Gao et al., 2015) to reduce the dimensions of the image representation, which enables quick processing in practical settings. DTR encoding was applied to 8,736 hematoxylin and eosin

(H&E)-stained histopathology whole-slide images (WSIs) of 32 cancer types from The Cancer Genome Atlas (TCGA) (Weinstein et al., 2013) (Table S1) and H&E-stained WSIs of biopsies from patients with gastric cancer who received PD-1 blockade therapy and then validated in three clinically useful applications: (1) unsupervised analysis of cancer morphology to detect histological subtypes in cancer, (2) content-based image retrieval (CBIR), i.e., the identification of histologically similar images among large datasets, and (3) prediction of genomic features (e.g., driver gene mutations) from histology images. After the selection of uniform cancerous regions by trained pathologists, 271,700 patches of 128 $\times$ 128 μm were extracted and resized to 256 $\times$ 256 pixels for further analysis.

RESULTS

Finding an optimal deep neural network for DTRs of cancer histology

The architecture of the neural network and the layers used within bilinear operations affects extracted features. Thus, we

first determined the optimal combination of the network and layer for DTRs—which properly captures characteristics of histopathology images—by evaluating the correlations between similarity rankings of the images calculated using features obtained by DTRs from various representative pretrained models, and the similarities were ranked by trained pathologists (Figures 1C and S1A; STAR Methods). We found DTRs extracted from the middle layers in a NasNet-A Large network best correlated with pathologist analyses among the several networks examined. In most CNN architectures, the middle layers outperformed the last layers within the same network, suggesting that as networks are trained using general images and convolutional layers learn spatial pattern hierarchies, middle layers that capture intermediate-level features are better suited to histopathology image analysis than the last layers that learn high-level features in larger receptive fields (Yosinski et al., 2015). We also evaluated the robustness of this result to the selection of image patches by exchanging the training patches with others randomly selected from the same regions, which are likely to have nearly identical histology (Figure S1B). We observed a similar overall tendency and that normal_concat_11 from the NasNet-A large network ranked top in terms of median correlation among 1,000 trials.

Next, a systematic comparison was performed to identify factors contributing to the performance of DTRs including feature-processing methods (global average pooling or DTRs), pretraining methods (supervised learning on ImageNet or supervised/self-supervised learning [He et al., 2020] on 271,700 patches from TCGA pathology images), and DTR dimensions (Figures S1C–S1D). We selected the VGG16 network for our comparison because the NASNet-A Large network was difficult to fine tune given its high-depth structure. Our results suggested the following: (1) DTRs perform better than another spatially orderless deep features using global average pooling, (2) the ImageNet pretrained model performs better than the model pretrained on TCGA histology images, which indicates that a network trained on a larger number of diverse objects better captures the histology features; and (3) 1,024-dimensional DTRs are comparable to higher dimensional DTRs.

We also compared DTR performance from the NASNet-A Large network using 1,024, 8,192 and 32,768 dimensions for final representations, and, as a result 1,024-dimensional DTRs exhibited comparable performances to higher dimensional DTRs in the NASNet-A network as well (Figure S1E), which suggests that the 1,024 dimension is preferable in order to reduce computational time and required memory. These results collectively suggest using the 1,024-dimensional DTRs from the ImageNet pretrained model as the representation in subsequent analyses considering the computation time.

Finally, the robustness of DTRs to brightness, contrast, sharpness, and color variation was also evaluated. The cosine similarities between the DTRs for original and augmented images were compared with those between the DTRs for original images and images extracted from the same annotated region, which are expected to have a similar texture (Figures S1F–S1G). The cosine similarity to the disturbed images was higher than that from the same annotated region, which demonstrates the robustness of the DTR features to the different variations.

Unsupervised pan-cancer histological profiling and subtyping by DTR

When 7,175 TCGA pan-cancer cases are projected in 2D space using their DTRs, we found that they are well distributed based on morphologic features and that histologically similar images came in close proximity (Figure 2A). While morphologically distinct histology images tend to form a distinct cluster, such as lower-grade glioma (LGG)/glioblastoma multiforme (GBM) or kidney renal clear-cell carcinoma (KIRC), those of different tissue origins with similar histopathological appearances (e.g., squamous cell carcinomas from multiple organs) tended to overlap significantly within each category (Figures 2B and S2A). Importantly, patches with similar morphology but differing H&E stain intensity also tended to fall close to each other, indicating that DTR encoding is robust despite color variation, which is an issue that frequently poses a major problem when comparing cases from different hospitals (Figure S2B). We also calculated the pairwise cosine similarity within individual, between individual, and between cancer type for each cancer type (Figure S3). The results showed the following. (1) The within-individual similarity was always highest. This indicates that tumors from the same patient share similar morphology and that DTRs successfully captured this. (2) The between-individual similarity ranked second to within-individual similarity in most cancer types, indicating that the morphologies from the same cancer type share similar properties, and DTRs captured cancer-type-specific features. In some cases, however, the similarity was low, and the median could be even lower than the between-cancer-type similarity for some tumors, probably because the tumor tissue is highly heterogeneous even in the same cancer type. (3) The between-cancer-type similarity is higher when the target cancers share common histological characteristics (e.g., adenocarcinoma, squamous cell carcinoma, and brain tumor), as suggested in Figures 2 and S2A.

DTRs also revealed the morphological diversity within a certain cancer type, which enables the identification of histological subtypes, including those that are rare. As exemplified by the Gleason grade in prostate adenocarcinoma and stomach intestinal/diffuse adenocarcinoma (Figure 3A), DTRs clearly showed differential distributions according to histological grade and subtypes within each cancer category, whereas there was no correlation with cell density (Figures S2C–S2D). Patches from a G4 case projected on the edge of the distribution and far from the other G4 cases exhibit a rare hypernephromatoid pattern, showing that histologically distinct subtypes are successfully mapped on different positions (Figure S2E). Additionally, DTRs of prostate cancer from the independent validation cohort (Gleason 2019 dataset [Nagpal et al., 2018]) also showed differential distributions according to the Gleason grade (Figure S2F). In pancreatic adenocarcinoma (PAAD), most cases were projected close to each other in a pan-cancer uniform manifold approximation and projection (UMAP) space, while there were some cases projected far from the majority (Figure 3B). Gene expression profiling revealed that these outlying cases, although initially annotated as PAAD without specific comments, exhibited different lineage differentiation, such as intestinal, neuroendocrine, and squamous differentiation, which were also confirmed by histological examination (Figures 3B and S2G). Notably, the cases

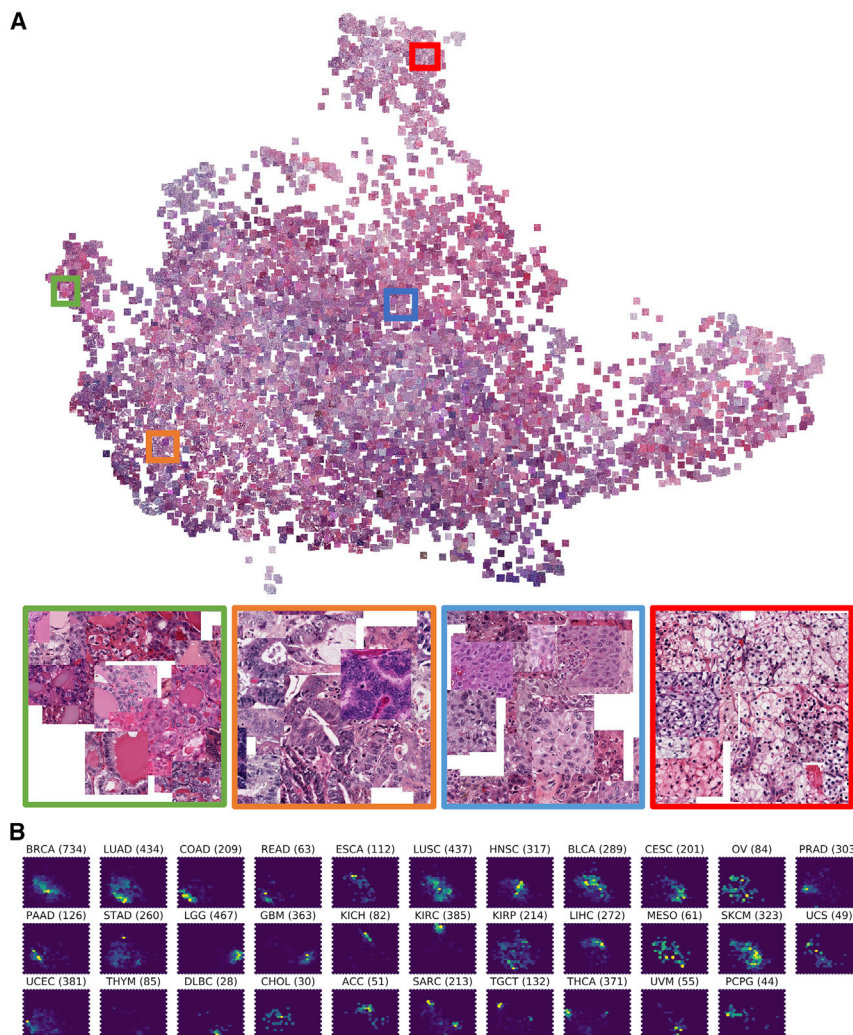


Figure 2. Two-dimensional projection of the DTRs of TCGA pan-cancer histology images

(A) Representative images ($n = 7,175$) ($128 \times 128 \mu\text{m}$) are shown at relative positions according to their DTR vectors using uniform manifold approximation and projection (UMAP). Enlarged images of the regions indicated by colored boxes are shown at the bottom.

(B) Distribution of each cancer type in the UMAP projection. The numbers in parentheses following the cancer type indicate the number of patients included.

LUAD, lung adenocarcinoma; COAD, colon adenocarcinoma; READ, rectum adenocarcinoma; PAAD, pancreas adenocarcinoma; LUSC, lung squamous cell carcinoma; HNSC, head and neck squamous cell carcinoma; CESC, cervical squamous cell carcinoma.

high expression levels of variable lineage-specific markers, such as prostate stem cell antigen (*PSCA*; prostate, stomach), *PDHA2* (testis), and *MUC21* (esophagus) (Figure S2I), and was also differentially projected from the other cases.

Unsupervised analysis of DTRs was also performed on another cohort of 35 patients with gastric cancer who were treated with the immune checkpoint inhibitor (ICI) nivolumab (Figures 3C–3G and S4; Table S2). Principal-component analysis of DTRs from tumor regions of biopsy specimens at the time of diagnosis revealed that the first principal component (PC1) correlated with the standard Lauren classification (intestinal or diffuse) as expected (Figure S4A). Notably, PC2 significantly correlated with the response

showing intestinal differentiation (I1: TCGA-US-A776 and I2: TCGA-HZ-7289) were closer to the colorectal cancer cases. These cases harbor *GNAS* mutations and are thought to be derived from an intraductal papillary mucinous neoplasm (IPMN) of the intestinal type, which correlates with the findings of another study (Molin et al., 2013). Exceptionally, two cases with a high squamous expression signature (TCGA-FB-A5VM) or a high neuroendocrine expression signature (TCGA-2L-AAQM) were around the center of the PAAD distribution. However, the diagnostic slide of the former did not contain squamous cell components, which was probably due to histological cancer heterogeneity and a different sampling locus used for RNA sequencing (RNA-seq) and histological analysis. The latter case has a significant artifact that shows severe detachment of the epithelium from the basement membrane and, coupled with its original trabecular histological pattern, resembles the gland structure of a well-differentiated ductal adenocarcinoma, possibly shifting the case toward the center of the PAAD distribution (Figure S2H). A case with a clear-cell-type component (C6: TCGA-3A-A9IU) showed unusual differentiation, with

to nivolumab treatment (Figures 3C and 3E; $p = 0.004375$, Jonckheere-Terpstra trend test) and the overall survival (OS) and progression-free survival (PFS) of these patients ($p = 0.0038$ for PFS, $p = 0.029$ for OS; Figure 3D; Table S3 for multivariate Cox models) despite the fact that the biopsies were sampled prior to the treatment. Immunohistochemistry staining for PD-L1 and CD8 indicated that the status of these markers was only weakly correlated with the response (PD-L1: $p = 0.252$ for Fisher's exact test and $\phi = 0.22$; CD8: $p = 0.267$ for Fisher's exact test and $\phi = 0.22$; Figures 3F and S4B). Because only one case with high microsatellite instability (MSI-H) and one Epstein-Barr virus (EBV)-infected case (Figure 3C) were included, which are known to be effective treatments for ICIs (Kim et al., 2018, p. 1), PC2 stratified the responses of usual-type microsatellite-stable (MSS) cases, in which the relationship to the response has not yet been well studied. Detailed histological analysis showed that PC2 positively correlated with cases that are considered indeterminate by Lauren classification, but PC2 was more strongly correlated with the response to nivolumab treatment (indeterminate versus intestinal or diffuse: $p = 0.0498$ for Fisher's

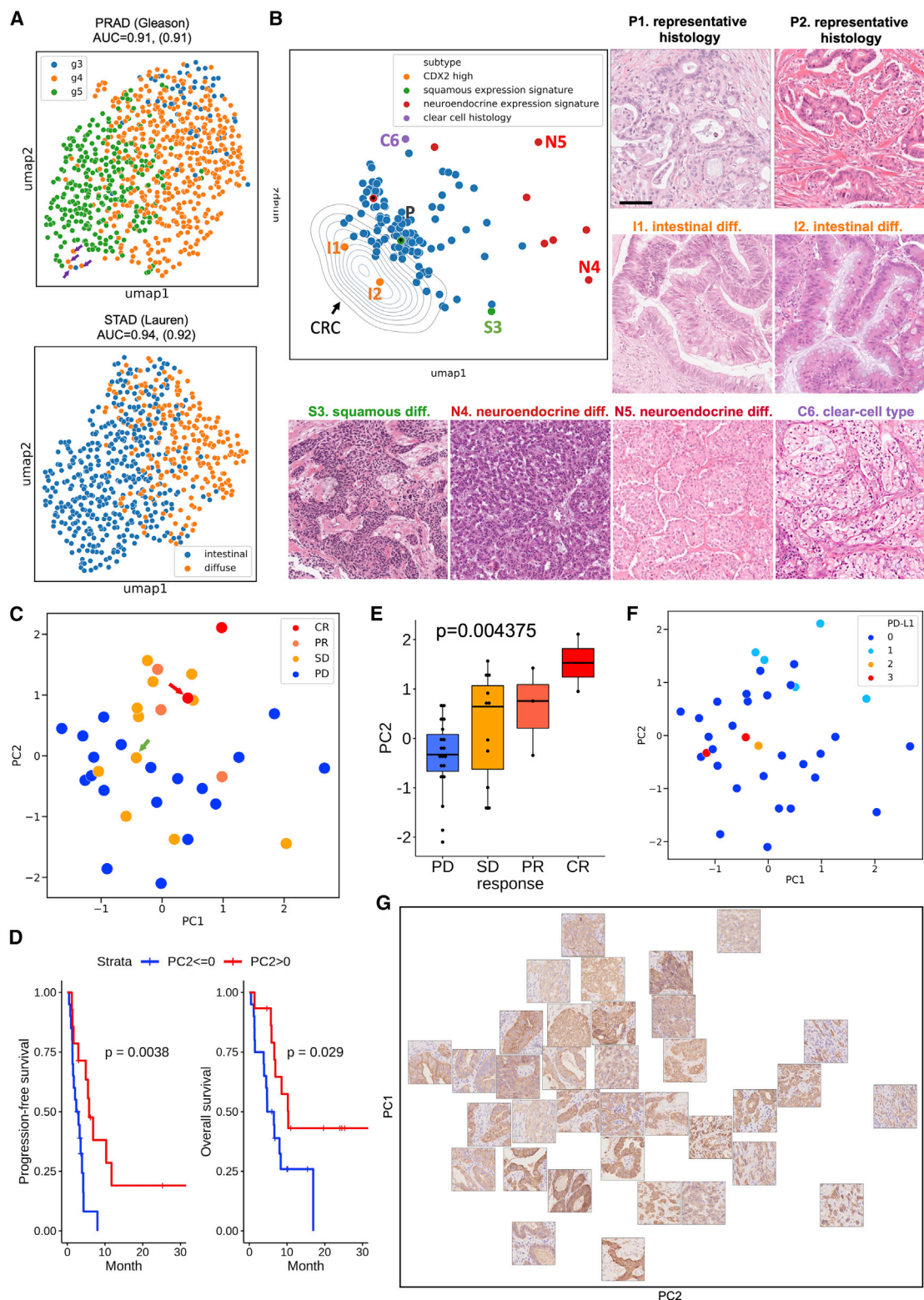


Figure 3. Unsupervised histological profiling and subtyping by DTRs

(A) UMAP projection of Gleason scores in prostate adenocarcinoma (top, n = 830) and Lauren classifications in stomach adenocarcinoma (bottom, n = 661) for each region selected by the pathologists. Mean area under the curve (AUC) values based on logistic regression using DTRs without or with (indicated in

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exact test and $\phi = 0.313$, PC2: $p = 0.0002$ for Fisher's exact test and $\phi = 0.576$; [Figures 3C and S4A](#)) than the histological classification. Cytokeratin immunostaining of these specimens indicated that tumor cells from cases with a good response form a multilayered cell nest, while those with a poor response form monolayer tubules or isolated small-cell aggregates ([Figures 3G and S4C–S4E](#)), indicating that cases showing tumor cells in close contact with stromal components are less responsive to ICIs. These results suggest that multilayered cell nests reflect the protection mechanism of highly immunogenic cells from immune cell attack, while monolayer tubules or isolated small-cell aggregates reflect that these tumor cells are not immunogenic. It has been suggested that the MSI-H status of colorectal cancer positively correlates with solid morphology or the production of mucinous stroma around tumor cells ([Alexander et al., 2001](#)). Collectively, some of the features extracted by DTR correlated with the response of ICI treatment from H&E histology images alone before treatment based on the unreported feature of the tumor morphology; however, further validation with more patients is necessary.

These results demonstrated that unsupervised histological profiling by DTR-based universal encoding comprehensively captures the diversity of cancer morphology and is useful for characterizing known and unknown subtypes.

CBIR of histologically similar images by DTR

There has long been a need to utilize CBIR to retrieve histologically similar images for educational, diagnostic, and research purposes, and this technology could contribute significantly to the standardization of diagnostic quality. We attempted CBIR by retrieving the most relevant images from TCGA pan-cancer datasets based on the cosine similarity of DTR spaces and the approximate nearest neighbor search method ([Figure 1A](#)). Using representative query images from multiple cancer categories in TCGA and independent validation datasets, we found that the retrieved images appeared histologically similar to the query images ([Figures 4A and S5A](#)). DTRs contributed to successful image retrieval for each query image from the same categories of 32 cancer types as defined by TCGA (precision: $90.8\% \pm 10\%$

among the top 10 ranked images, and $78.8\% \pm 15.4\%$ among the top 20 ranked images) ([Figures 4B and S5B](#)). To further test whether encoding by DTRs is capable of retrieving images that are histopathologically similar, we then performed a strictly blinded comparison to evaluate the concordance between the cosine similarity defined by DTRs and the histopathological similarity judged by 5 independently trained pathologists. As a result, images that the CBIR determined to be most similar to the query images and those selected by five pathologists were almost identical (59/60 tests), except for one case where the image judged to be the second-most similar by the DTR was selected by one of the pathologists to be the first-most similar ([Figure S6](#)). These results demonstrated that DTRs effectively capture histopathological features and CBIR using this encoding method at a practical level in clinical settings.

Comprehensive evaluation of cancer-genome-histology correlations by the DTRs of H&E image

Because histology encoding by DTRs does not require extra training on the target images; once they are generated, they can be correlated to diverse biological and clinical parameters using simple machine-learning models, such as logistic regression, which have less hyperparameters and are less prone to overfitting than other deep-neural-network-based models. We therefore examined the correlation between histology and cancer genome aberrations, aiming to discover potential morphologic features induced by somatic cancer driver mutations. The correlation between DTRs and 208 informative somatic mutations, including single-nucleotide variations (SNVs), insertions or deletions (indels), copy-number variations (CNVs), and fusions, out of 322 cancer driver genes was investigated in the OncoKB database ([Chakravarty et al., 2017](#)) using 1,024-dimensional DTRs as features for a logistic regression model. Stratified 3-fold cross validation was performed, and subsets of these data were validated in our independent datasets outside of TCGA (see [STAR Methods](#)). We discovered that 309 genomic-feature-cancer-type combinations, including 42 clinically actionable mutation statuses, exhibited significant correlations with DTRs and are predictable based solely on H&E histopathology images (area under the curve [AUC] of

parentheses) color normalization are also shown. Arrows within the Gleason score plot indicate a case with a Gleason score of 4 at outlier-like positions, which belong to a hypernephromatoid pattern subtype ([Figure S2E](#)). Color normalization did not improve classification performance.

(B) UMAP projection of PAAD in pan-cancer histology distribution ($n = 126$). Representative histology of pancreatic ductal adenocarcinoma around the center is shown (P1: TCGA-HV-A70L and P2: TCGA-IB-7889). Rare histological subtypes, including intestinal differentiation with high CDX2 expression (orange, I1: TCGA-US-A776 and I2: TCGA-HZ-7289), adenosquamous differentiation with high squamous expression signature (green, S3: TCGA-2J-AAB1), neuroendocrine differentiation with high neuroendocrine expression signature (red, N4: TCGA-3A-A91L and N5: TCGA-3A-A91J), and clear-cell-type carcinoma (purple, C6: TCGA-3A-A91U) are differentially mapped from the other cases. The two cases with intestinal differentiation were closer to colorectal cancer (COAD and READ) distribution, as shown by a contour. Cases with an asterisk (*) are exceptionally close to the center of the PAAD distribution despite exhibiting squamous or neuroendocrine expression signatures (see the main text). Scale bar: 100 μm . All images have the same scale.

(C) Principal-component analysis (PCA) of DTRs from gastric cancer biopsies at the time of diagnosis before nivolumab treatment ($n = 35$). Colors indicate the response to nivolumab treatment. Red and green arrows indicate MLH1-negative (MSI-H) and EB-virus-infected cases, respectively.

(D) Progression-free survival (left) and overall survival (right) after the nivolumab treatment of patients with gastric cancer. Patients were stratified based on the PC2 value (>0 or ≤ 0). The p values for the two-sided log-rank test are shown.

(E) PC2 score distributions for the response to nivolumab treatment. The p value for the two-sided Jonckheere-Terpstra trend test is shown. PD, progressive disease ($n = 19$), SD, stable disease ($n = 11$), PR, partial response ($n = 3$), CR, complete response ($n = 2$). For the box plot, the lower and upper hinges correspond to the 25th and 75th percentiles, respectively; the upper whisker extends from the hinge to the largest value no further than $1.5 \times$ interquartile range (IQR) from the hinge. The lower whisker extends from the hinge to the smallest value at most $1.5 \times$ IQR of the hinge.

(F) PD-L1 IHC score distribution on the PCA plot.

(G) Representative AE1/3 (cytokeratin)-stained histological images of each sample placed on the PCA plot. The positions are manually adjusted so that the histology images do not overlap with each other.

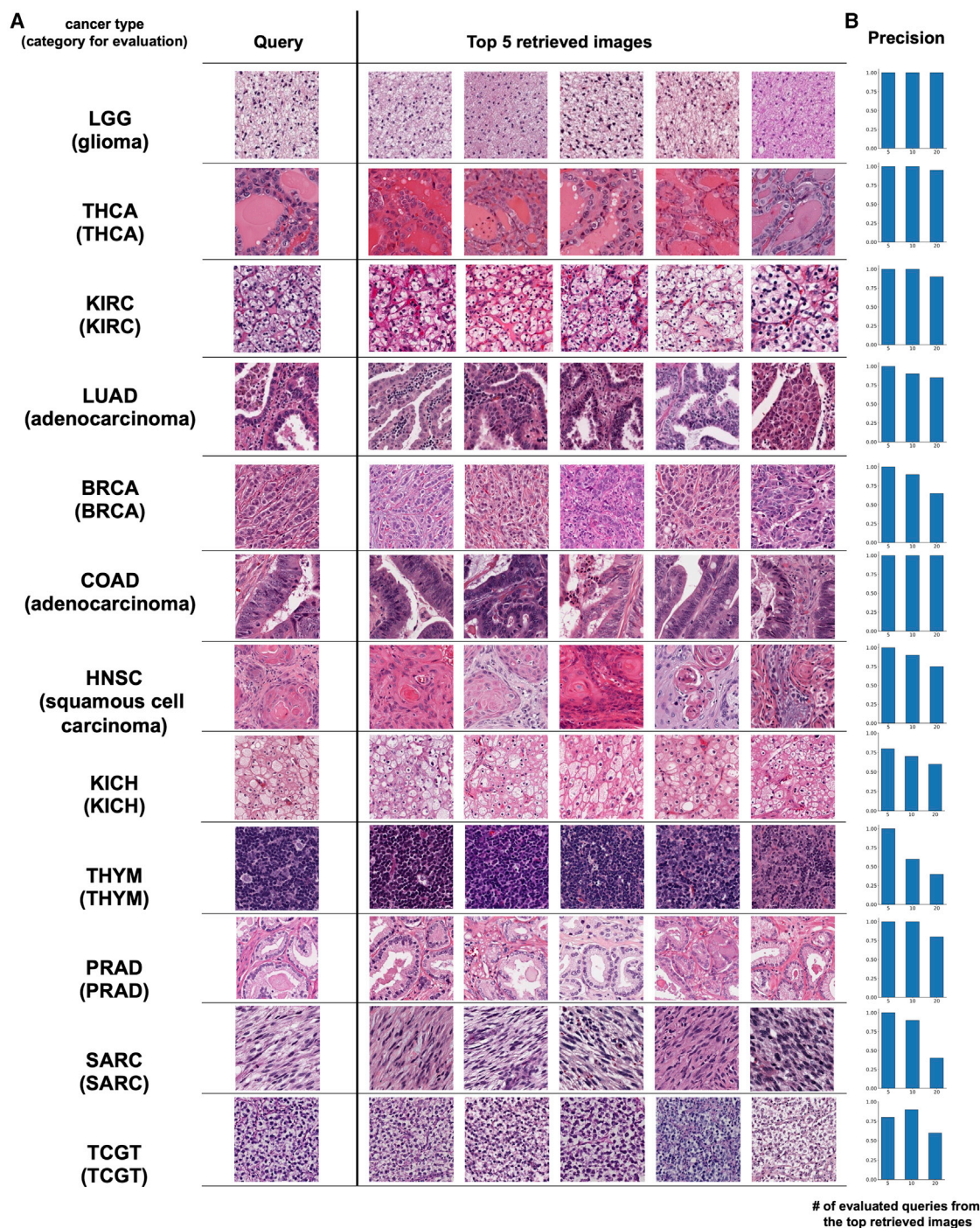


Figure 4. Content-based image retrieval of histologically similar images by DTR

(A) CBIR extraction of histologically similar images. Representative query images and the top 5 retrieved results are shown for 12 cancer types. The five images were retrieved from the cancer histology images of 6,535 patients (640 patients used for the DTR network and layer selection were excluded) across all cancer types in TCGA dataset based on the cosine similarity by DTRs. Patch sizes of the query and the retrieved images are $128 \times 128 \sim 256 \times 256 \mu\text{m}$.

(B) Precision (number of images in the correct category/number of retrieved images) of CBIR for the 12 query images. Precision values based on the categories for the top 5, 10, and 20 retrieved histopathology images are shown. The cancer type was used as the category except for LUAD, COAD, and HNSC, in which adenocarcinoma or squamous cell carcinoma were used as the ground-truth category.

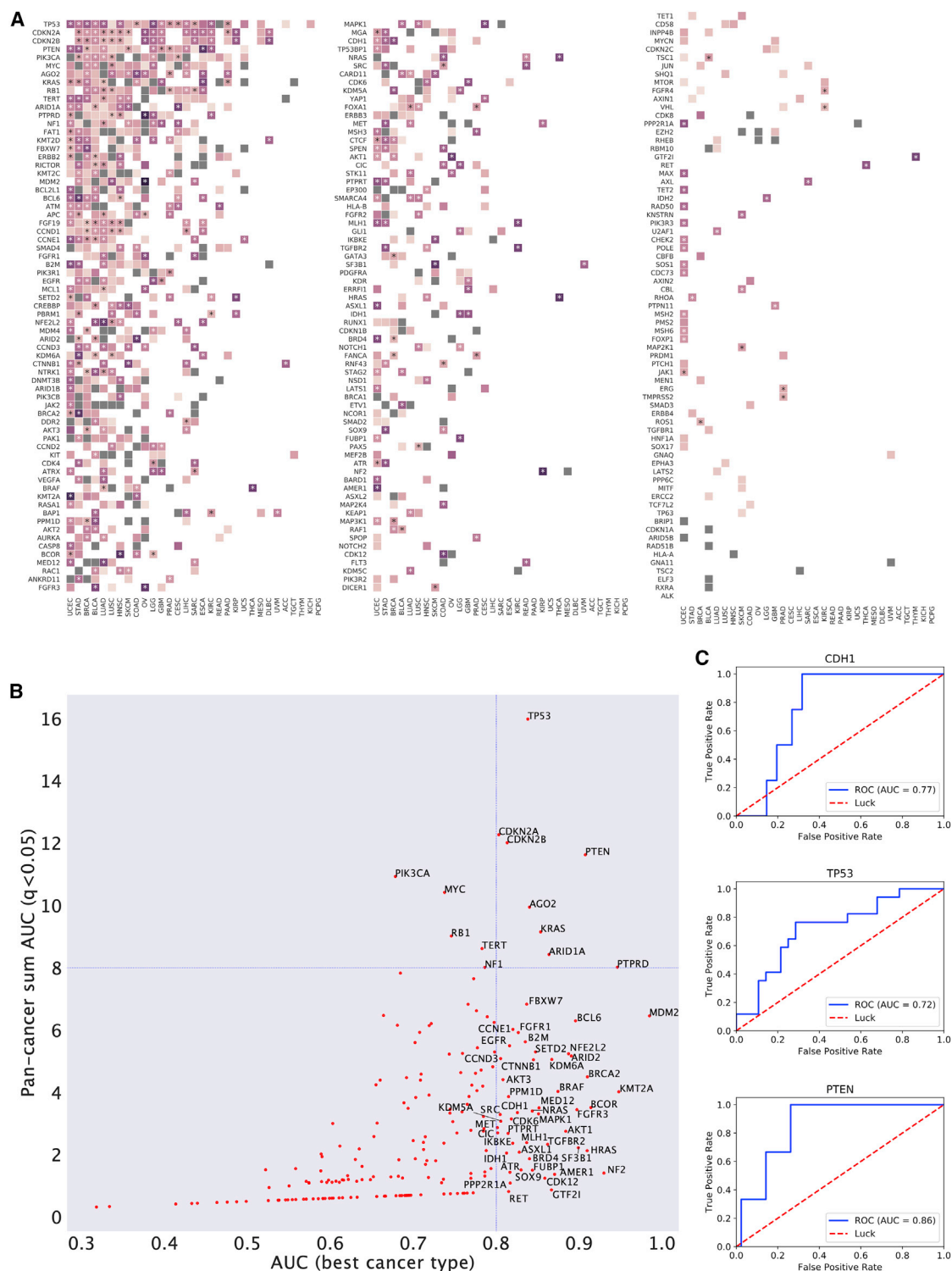


Figure 5. Prediction of cancer-related genomic features from the DTRs of H&E histology image

(A) Area under the receiver operating characteristic (ROC) curve values for the prediction of somatic mutations of OncoKB-annotated 208 driver genes, and somatic SNVs, indels, CNVs, and fusions are integrated in that their direction of gain- or loss-of-function does not contradict each other. Asterisks indicate $q < 0.02$; gray boxes indicate $AUC < 0.5$; white space indicates that the mutations could not be predicted due to a lack of an adequate sample number for genomic aberrations. The number of cases is summarized in Table S4.

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receiver operating characteristics >0.70 , $q < 0.02$; [Figures 5A and 5B](#) and [S7A–7D](#); [Tables S4 and S5](#)). These include previously reported correlations, such as STK11 and TP53 in lung adenocarcinoma (LUAD) ([Coudray et al., 2018](#)) and SPOP in prostate adenocarcinoma (PRAD) ([Schaumberg et al., 2017](#)) as well as those coupled with known subclasses, such as CDH1 mutations linked to lobular breast carcinoma and diffuse-type gastric adenocarcinomas ([The Cancer Genome Atlas Research Network et al., 2014](#); [Pharoah et al., 2001](#)). When our analysis is limited to SNV/indels, the prediction accuracy of the DTR-based classifier in terms of AUC is significantly higher than recently published results ([Kather et al., 2020](#)) using trained CNN models ($p = 7.323 \times 10^{-9}$, 95% confidence interval [CI] of difference: 0.04–0.08, paired t test; [Table S6](#)), although factors other than the classifier, such as patient or patch selection, could contribute to the difference in performance. Additionally, our analysis covers much broader cancer types and drivers, including CNVs, and most of the correlations found in this study have not been previously described in detail. Uterine corpus endometrial carcinoma (UCEC) and stomach adenocarcinoma (STAD) contain morphologic correlations with many driver genes, thereby reflecting their major distinct biological subtypes (i.e., endometrioid/serous/MSI in UCEC and diffuse/intestinal/MSI/EBV⁺ in STAD). While some driver gene mutations are linked to the morphology of a specific cancer type, such as *GTF2* to thymoma (THYM) and *NF2* to kidney renal papillary cell carcinoma (KIRP), general cancer driver genes, such as TP53, CDKN2A/CDKN2B, PIK3CA, PTEN, and MYC, which are responsible for well-known cancer hallmarks, including genomic stability, cell cycle, and cellular metabolism ([Hanahan and Weinberg, 2011](#)), are linked to histologic features in a broad range of cancer types ([Figures 5A and 5B](#)). The performance of models trained on TCGA cases for the prediction of driver mutations is validated by our independent STAD datasets ([Figure 5C](#)).

Correlation of morphological features with somatic genomic mutations

We then investigated what morphological changes are linked to these driver gene mutations and mutation signatures. A comparison of top- and bottom-scored images for individual driver genes revealed distinct morphologic characteristics in mutant-positive cases ([Figure 6A](#) and [S8A–S8C](#)), which were even recognizable upon visual inspection by non-pathologists. Some of these features can also be expressed with classical histopathologic keywords ([Table S7](#)); for example, FGFR3-mutated bladder urothelial carcinoma (BLCA) exhibits regularly arranged cells with thin vascular stroma, and CTNNB1-mutated skin cutaneous melanoma (SKCM) exhibits sheet-like spindle cell tumors with clear melanin pigment production ([Figure S8A](#)). By examining morphological changes induced by broad effectors in multiple cancer types, we have found that some oncogenic drivers induce universal morphologic change across cancer types. For example, the TP53 mutation—which causes chromosomal instability—induces an increase in nuclear size and atypia within

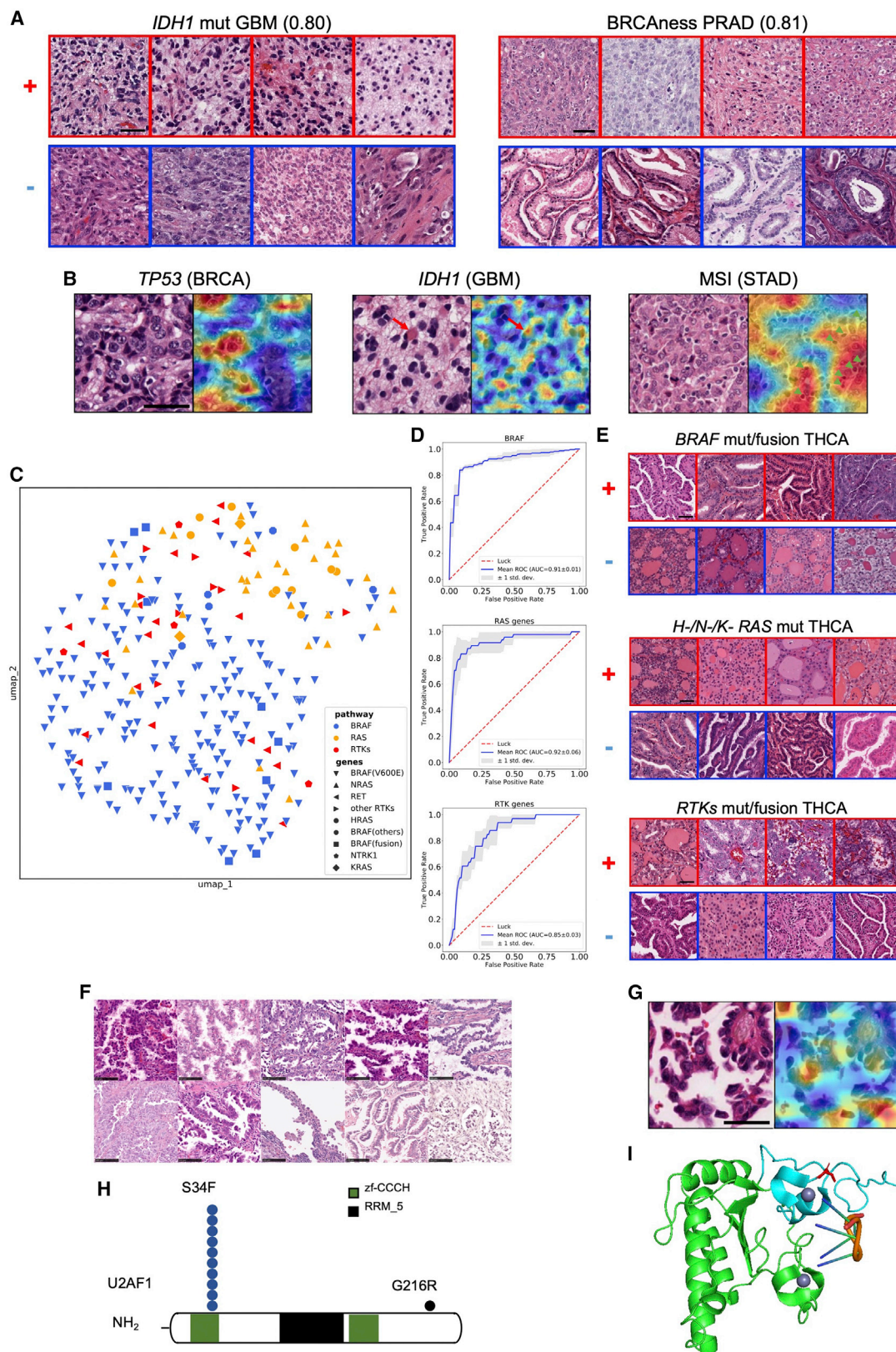
LUAD, breast-invasive carcinoma (BRCA), UCEC, head and neck squamous cell carcinoma (HNSC), and GBM ([Figure S8B](#)). In contrast, MYC amplification induces an increase in cell size with plump cytoplasm, typically in liver hepatocellular carcinoma (LIHC), PAAD, lung squamous cell carcinoma (LUSC), BRCA, and BLCA ([Figure S8C](#)). As MYC is a major accelerator of glycolytic metabolism, our result indicates a relationship between cell-size regulation and cellular metabolism, which has also been suggested by several lines of experimental evidence ([Baena et al., 2005](#); [Zanet et al., 2005](#)).

In some cancer-type-driver gene combinations with large sample sizes and appropriate class balances that were fed into the learning of the deep neural network, we obtained an objective depiction of the biologically reasonable, mutant-specific, morphologic features using the Grad-CAM approach ([Figures 6B and S8D](#); see [STAR Methods](#)). These results suggest that while TP53 mutations in breast cancer are linked to atypical cancer nuclei, IDH1 mutations in GBM are not linked to nuclear features but to astrocyte-like faint and/or gemistocytic cytoplasmic features, possibly reflecting the development of IDH1-mutant GBM from low-grade astrocytoma ([Cancer Genome Atlas Research Network et al., 2015](#)). Mutation signatures associated with MSI in STAD could overlap with intraepithelial lymphocytic infiltration, potentially reflecting the well-established link between neoantigens and intratumor lymphocytic infiltration ([Phillips et al., 2004](#)). Within thyroid carcinoma (THCA), we observed a clear link between oncogenic pathway mutations and specific morphology; BRAF, RAS (HRAS and NRAS), and RTK (*RET*, NTRKs, *ALK*, etc.) mutant groups exhibited group-specific morphologic features, indicating that similar pathway signals generate similar tumor morphology and biology ([Figures 6C–6E](#)).

Through these studies, we found distinct and unreported subgroups with both characteristic genomic and morphologic features that would not be easily found by simply grouping with different mutations. U2AF1 is a component of a spliceosomal complex, and LUAD with a U2AF1 mutation is a well-differentiated adenocarcinoma exhibiting distinct lepidic predominant tumors with a hobnail appearance of the tumor cells ([Figures 6F–6I](#) and [S8E](#)). Almost all cases (10/11) had an S34F hotspot mutation, suggesting a gain-of-function nature for the mutant proteins. Serine at the 34th amino acid, which is in the conserved zf-CCCH domain and close to the target RNA, is highly conserved across species. CTNNB1 is a transcription factor complex involved in adrenal development ([Berthon et al., 2012](#)), and CTNNB1-mutated adrenocortical carcinoma (ACC) comprises regularly packed cells with round nuclei ([Figure S8F](#)). All CTNNB1 mutations found in this subgroup were at the mutation hotspot in exon 3. Although pathologists can note the correlation of morphology and specific genomic features without difficulty once presented, quantitative encoding of cancer histology makes it practically feasible to survey all possible combinations of mutations and cancer types comprehensively.

(B) A global view of pan-cancer and cancer-type-specific correlations between morphology and genomic aberrations. x and y axes represent AUC values and total AUC values with $q < 0.05$ in the 32 cancer types, respectively.

(C) Performance of the logistic regression models using DTRs trained with TCGA STAD datasets on our independent STAD cohort ($n = 46$) was measured in terms of AUC for *CDH1* (4 positive cases), *TP53* (17 positive cases), and *PTEN* mutations (3 positive cases).



(legend on next page)

DISCUSSION

In this study, we demonstrated universal encoding of diverse cancer histology images using DTRs and its usefulness in three clinically important applications. A small number of previous researchers applied B-CNN to histopathology images, but they were limited to the supervised classification of histological images into predefined classes (Wang et al., 2017; Liu et al., 2020). In this study, we have revealed that the features extracted by the universal process using pretrained B-CNN from optimized combinations of network models and layers well express complex pan-cancer histology and are practical in clinical settings beyond the classification of a certain cancer type.

The histology images are different from the general images when we see the whole image, but we hypothesized that the small part of the images, which can be captured by the early-to-mid-layers of the CNN, are similar. That was partially proved in Figure 1C, which shows that the DTRs extracted from mid-layers were well correlated with pathologists' intuition. Additionally, the ImageNet pretrained model was trained on much more diverse images than the histology images, and it could get more various mid-level image features that could be robust to color variations. We think that is one of the reasons why our model outperformed the model trained on TCGA images, as shown in Figures S1C–S1D. Although it could be probable that the models trained on much larger amounts of diverse histology images from many hospitals surpass our model, our model performs sufficiently well, as shown in the series of experiments.

Once the DTRs are extracted, they can be dealt with as well-structured data, such as gene-expression profiles. As gene-expression profiling is a common approach to characterize tumor diversities and subtypes, unsupervised DTR analysis can be a common approach to characterize tumor diversities from histopathology and to subtype them by including the dis-

covery of rare subgroups, which was exemplified by our analysis of PAAD and other cases in this study. Additionally, unsupervised analysis of DTRs on a cohort of patients with gastric cancer demonstrated that the response of ICI treatment can be predicted from H&E biopsy images alone even before treatment, based on the previously unrecognized feature of the tumor morphology. As most pathologists manually refer to a histology atlas when encountering unknown cases, CBIR would drastically decrease the time required for a correct diagnosis. This is especially useful in small- to medium-sized hospitals and those in developing countries, where pathologists often work alone without any practical access to consults. Our studies also uncovered 42 clinically actionable mutations and genomic features that are predictable with high accuracy (AUC >0.70) from H&E images, including EGFR, ERBB2, FGFR1/2/3, PIK3CA, IDH1/2, BRAF, KIT, and BRCAness as well as MSI status (Figures S7C–S7D). Although this method is not 100% accurate and requires genetic testing for a definitive diagnosis, the systematic selection of highly probable candidates is important in many contexts. Particularly in developing countries where clinical resources are limited, it could lead to a substantial reduction of negative results and the resultant total number of subsequent costly cancer gene panel tests at the population level.

To facilitate DTR-based analyses for routine H&E histologic images, we implemented our program both to a web server (Figure 7A) and smartphone devices (Figures 7B and 7C; Video S1) as “Luigi” (Large-Scale Histopathological Image Retrieval System). By using a smartphone microscope device, clinicians not equipped with a digital slide camera device can take images and access our developed image analysis program. An effective dimensionality reduction of DTRs successfully enabled a “query-to-display time” of less than 4 s, which is practical in routine clinical settings. Our studies have unveiled the link between common actionable mutations and DTRs in major cancers

Figure 6. Morphologic features correlated with somatic genomic mutations and a biologically distinct subclass, as discovered by DTR-based comprehensive evaluation of cancer histology

(A) Representative images of cases with high (top row) and low (bottom row) somatic mutation probability based on logistic regression models are shown for IDH1 mutations in GBM, and BRCAness signature in PRAD. Images with high and low probabilities were selected from mutation-positive and -negative cases, respectively. The frame color indicates the presence (red) or absence (blue) of mutations. All eight images shown for each gene category are of the same magnification. Numbers in parentheses indicate mean AUCs of the prediction for each cancer-mutation combination in 3-fold cross validation. Scale bar: 50 μ m. All images in each mutation have the same scale.

(B) Gradient-weighted class activation mapping (Grad-CAM) visualization of mutation-related morphological changes by CNN. These results suggest that TP53 and IDH1 mutations are primarily linked with the nucleus and cytoplasm of cancer cells, respectively, and that the MSI status overlapped with tumor-infiltrating lymphocytes. Arrows show a gemistocytic cytoplasmic feature in GBM, and arrowheads show tumor-infiltrating lymphocytes (TILs) in the MSI (STAD). Scale bar: 50 μ m. All images have the same scale.

(C) Distribution of histology images of THCA cases within UMAP projection (n = 371). Three pathway mutations are indicated by the different colors: BRAF, H-/N-/KRAS, and receptor tyrosine kinases (RTKs) (RET, NTRKs, and ALK). The dot shape indicates the mutated gene in each case. BRAF- and X-RAS-mutated cases are differentially positioned, and many RTK-mutated cases fall in the intermediate zone.

(D) The cross-validation performance of the logistic regression models using DTRs was measured in terms of AUCs for BRAF (223 positive cases), RAS (NRAS/HRAS/KRAS, 47 positive cases), and RTK mutations (33 positive cases). AUC values in RASs and RTKs were larger when measured altogether versus individually (0.91 for HRAS and 0.89 for NRAS; 0.77 for RET and 0.76 for NTRK1).

(E) Papillary structure with tall cells is observed in BRAF-mutated cases, while HRAS- and NRAS-mutated cases showed follicular morphology. RTK-mutated cases exhibit structures in which the epithelial front and back are reversed. All images for each gene-cancer combination are shown at the same magnification. Both somatic SNV and fusion were taken into account in (D) and (E). Scale bar: 50 μ m. All images in each mutation have the same scale.

(F–I) A distinct lung adenocarcinoma subclass with splice-complex protein U2AF1 mutations. (F) U2AF1-mutated cases share a similar histology of differentiated lepidic-predominant adenocarcinoma with hobnail-type cells. Scale bar: 100 μ m.

(G) Grad-CAM visualization potentially highlights the hobnail appearance of epithelial projections as a feature linked to U2AF1 mutations. Scale bar: 50 μ m.

(H) All cases, with one exception, had an S34F hotspot mutation on the CCH zinc finger domain, which strongly suggests its driver nature in this subgroup.

(I) The 3D position of S34, which is highly conserved across species, is displayed using the crystal structure of the U2AF1 complex (PDB: 7C07). The red-colored S34 of U2AF1 (green) is in the conserved zf-CCH domain (marine blue), close to the target RNA.

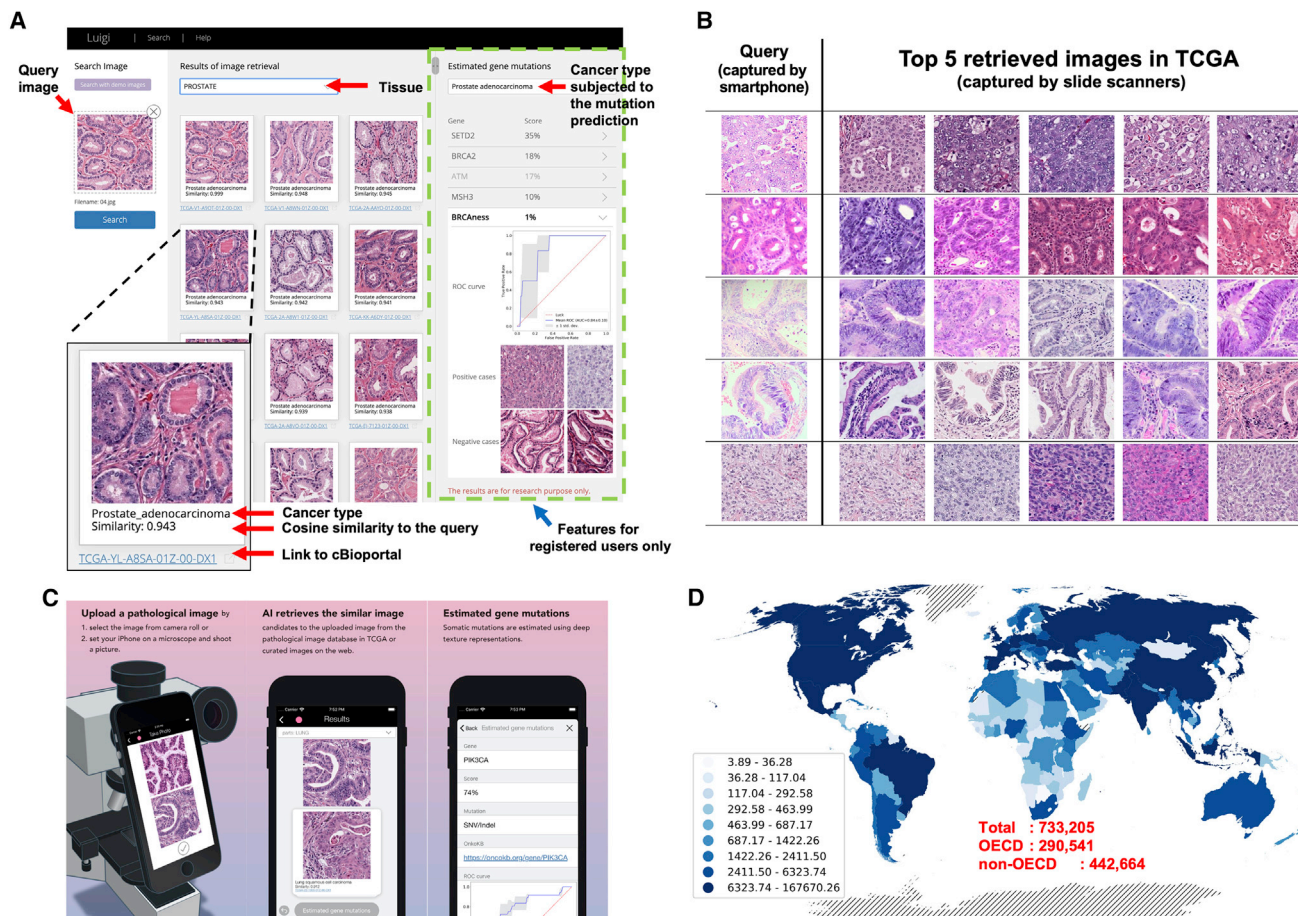


Figure 7. Global impact of CBIR and actionable mutation prediction with its implementation in an easily accessible environment

(A) Users upload a histopathological image either by dragging and dropping the file, pasting a screenshot of the image, or clicking “click here to select a file.” The Luigi system will retrieve the most relevant patch in each WSI, rank each patch according to cosine similarity, and initially display the relevant cancer cases among all cancer types. Tissues of origin can be changed by selecting the tissue from the pull-down menu above the results. By clicking the TCGA slide ID below each image, users can display the summary of the selected case on the cBioPortal website (<https://www.cbioportal.org/>), which includes the pathology report as well as clinical and molecular information for the case. The right panel shows the predicted genomic aberrations.

(B) CBIR using query images acquired with a smartphone to extract histologically similar images. Examples of query images and the top 5 resulting retrieval results corresponding to each query image are shown. Patch sizes of the retrieved images are $128 \times 128 \sim 256 \times 256 \mu\text{m}$.

(C) User interface and usage of CBIR smartphone application. Users upload original cancer histology images captured using a smartphone camera, and histologically similar images and predicted mutations are displayed.

(D) A global map showing the estimation of annual cancer cases with clinically actionable mutations that can be identified with our procedure with $>70\%$ positive prediction value (see STAR Methods). The total number of cases found to be positive annually using our method in OECD and non-OECD countries is shown in red.

such as LUAD, BRCA, colon adenocarcinoma (COAD), PRAD, and STAD. Our rough estimation by interpolating the actionable mutation frequency in TCGA to global cancer statistics indicates that approximately 0.73 million cancer cases with positive actionable mutations can be identified with a positive prediction value of $>70\%$ each year, of which 0.44 million cases are from non-Organization for Economic Co-operation and Development (OECD) developing countries (Figures 7D and S9A; see STAR Methods). Although this is the theoretical maximum under the assumption that most cases have equal access to routine pathological evaluation, more than 40 cancer cases can be identified per one thousand patients within both OECD and non-OECD countries by the same criteria (Figure S9B). When the statistics

are confined to cases treatable with small-molecule drugs that can easily be stored and transported (e.g., BCRness signatures correlated with platinum drug efficacy), 0.56 million cases worldwide (of which 0.35 million are from non-OECD countries) can be identified at the maximum (Figures S9C and S9D; Table S8).

DTRs might also be used to analyze multiregional differences in genomic features and evolutionary timing of individual variants or to annotate variants of unknown significance from histology images. For the former application, it is possible if the differences in genomic features affect the morphology. The latter application is possible if the affected pathway can be predicted based on DTRs, as shown in the THCA example (Figure 6D). Although further validation is required, they are included in future work.

Although our study is limited to cancer histology, it is possible to extend this encoding method to the non-cancer fields as well. By acquiring a universal quantitative encoder for routinely used H&E histology, pathologists can collect, accumulate, analyze, and query cases, as is the case with usual-structured data, such as genomic information. This is a major driving force in transforming histopathology into data science and is thought to have a profound impact on biomedicine and global health.

Limitations of the study

There are some limitations in this work. First, despite the selection of network and layer for DTR analysis being robust for the choice of images, the selection is based on the annotated data comprised of only one case for each cancer type. Thus, there is still room to consider whether the selected model and layer would be robust in images from different cases. However, we note that DTRs do not overfit (see [STAR Methods](#) and [Figure S10](#)) and that the DTR performance is strong in practice, as shown in the many experiments presented in this paper. Second, we excluded slides with poor staining or those that were out of focus from the analysis; however, such images could still exist in clinical settings. Although we did not evaluate how they could affect the analysis, such issues could be resolved by restaining for poorly stained slides or by using computational methods to detect blur, such as those that use the Laplacian filter for blurred WSIs. Therefore, we believe that these issues will not seriously limit the clinical utility of DTRs. Third, there is the potential overlooking of the relationships between histology and genomic features. Although the classification performance of mutation prediction is superior to previous work, it is still possible that the low AUC score of a mutation does not necessarily mean a lack of relationship with pathological phenotype. Analysis with more cases with the target mutation could uncover the relationships between the mutation and the histology that were not detected in this work.

STAR★METHODS

Detailed methods are provided in the online version of this paper and include the following:

- [KEY RESOURCES TABLE](#)
- [RESOURCE AVAILABILITY](#)
 - Lead contact
 - Materials availability
 - Data and code availability
- [EXPERIMENTAL MODEL AND SUBJECT DETAILS](#)
 - Unsupervised DTR analysis for gastric cancer specimens treated with anti-PD-1 checkpoint inhibitor
 - External validation cohort of genomic aberration prediction
- [METHOD DETAILS](#)
 - TCGA whole slide image dataset
 - Image preprocessing of TCGA images
 - Deep texture representations (DTRs)
 - Selection of optimal neural network architecture and layer

- Evaluation of the robustness of DTRs to image perturbation
- UMAP visualization of DTRs
- Prediction of gleason score for PRAD and lauren classification for STAD
- Cell count estimate using Cellpose
- DTR distribution for the independent external validation cohort for prostate cancer
- Gene expression analysis to identify rare subtypes of pancreas adenocarcinoma
- Unsupervised DTR analysis for gastric cancer specimens treated with anti-PD-1 checkpoint inhibitor
- Color normalization
- Independent validation cohort for CBIR evaluation
- Genomic aberration and clinical data of TCGA
- Prediction of genomic features from histopathology images using DTRs and supervised machine learning
- Performance comparison of driver mutation prediction with ShuffleNet-based classifier
- External validation cohort of genomic aberration prediction
- Visualization using Grad-CAM
- Luigi system for CBIR and genomic aberration prediction
- Estimation of the number of patients with predictable actionable mutations

● [QUANTIFICATION AND STATISTICAL ANALYSIS](#)

SUPPLEMENTAL INFORMATION

Supplemental information can be found online at <https://doi.org/10.1016/j.celrep.2022.110424>.

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AUTHOR CONTRIBUTIONS

Conceptualization, S.I. and D.K.; methodology, K.F. and D.K.; software, D.K., K.T., R.S., H. Konishi, and T. Umezaki; validation, D.K., M.O., and T.O.; formal analysis, D.K. and K.S.; investigation, D.K. and S.I., H. Koda, G.F., S.N., H. Kato, and A.K.; resources, D.K., N.S., Y. Sato., H.Y., K.K., Y. Seto, T. Ushiku, and M.F.; data curation, H. Koda and A.K.; writing – original draft, D.K. and S.I.; writing – review & editing, D.K. and S.I.; visualization, D.K. and F.M.; supervision, S.I.

DECLARATION OF INTERESTS

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REFERENCES

- Abadi, M., Agarwal, A., Barham, P., Brevdo, E., Chen, Z., Citro, C., Corrado, G.S., Davis, A., Dean, J., Devin, M., et al. (2016). TensorFlow: large-scale machine learning on heterogeneous distributed systems, Preprint at. arXiv, 160304467 Cs.
- Alexander, J., Watanabe, T., Wu, T.T., Rashid, A., Li, S., and Hamilton, S.R. (2001). Histopathological identification of colon cancer with microsatellite instability. *Am. J. Pathol.* 158, 527–535. [https://doi.org/10.1016/S0002-9440\(10\)63994-6](https://doi.org/10.1016/S0002-9440(10)63994-6).
- Alexandrov, L.B., Kim, J., Haradhvala, N.J., Huang, M.N., Tian Ng, A.W., Wu, Y., Boot, A., Covington, K.R., Gordenin, D.A., Bergstrom, E.N., et al. (2020). The repertoire of mutational signatures in human cancer. *Nature* 578, 94–101. <https://doi.org/10.1038/s41586-020-1943-3>.
- Baena, E., Gandarillas, A., Vallespinós, M., Zanet, J., Bachs, O., Redondo, C., Fabregat, I., Martínez-A, C., and de Alborán, I.M. (2005). c-Myc regulates cell size and ploidy but is not essential for postnatal proliferation in liver. *Proc. Natl. Acad. Sci. U S A* 102, 7286–7291. <https://doi.org/10.1073/pnas.0409260102>.
- Balanis, N.G., Sheu, K.M., Eshedbe, F.N., Patel, S.J., Smith, B.A., Park, J.W., Alhani, S., Gomperts, B.N., Huang, J., Witte, O.N., and Graeber, T.G. (2019). Pan-cancer convergence to a small-cell neuroendocrine phenotype that shares susceptibilities with hematological malignancies. *Cancer Cell* 36, 17–34.e7. <https://doi.org/10.1016/j.ccell.2019.06.005>.
- Bejnordi, B.E., Veta, M., Diest, P.J. van., Ginneken, B. van., Karssemeijer, N., Litjens, G., van der Laak, J.A.W.M., Hermesen, M., Manson, Q.F., Balkenhol, M., et al. (2017). Diagnostic assessment of deep learning algorithms for detection of lymph node metastases in women with breast cancer. *JAMA* 318, 2199–2210. <https://doi.org/10.1001/jama.2017.14585>.
- Berthon, A., Martinez, A., Bertherat, J., and Val, P. (2012). Wnt/β-catenin signalling in adrenal physiology and tumour development. *Mol. Cell. Endocrinol* 351, 87–95. <https://doi.org/10.1016/j.mce.2011.09.009>.
- Cancer Genome Atlas Research Network; Brat, D.J., Verhaak, R.G.W., Aldape, K.D., Yung, W.K.A., Salama, S.R., Cooper, L.A.D., Rheinbay, E., Miller, C.R., Vitucci, M., Morozova, O., et al. (2015). Comprehensive, integrative genomic analysis of diffuse lower-grade gliomas. *N. Engl. J. Med.* 372, 2481–2498. <https://doi.org/10.1056/NEJMoa1402121>.
- Cerami, E., Gao, J., Dogrusoz, U., Gross, B.E., Sumer, S.O., Aksoy, B.A., Jacobsen, A., Byrne, C.J., Heuer, M.L., Larsson, E., et al. (2012). The cBio cancer genomics portal: an open platform for exploring multidimensional cancer genomics data. *Cancer Discov.* 2, 401–404. <https://doi.org/10.1158/2159-8290.CD-12-0095>.
- Chakravarty, D., Gao, J., Phillips, S., Kundra, R., Zhang, H., Wang, J., Rudolph, J.E., Yeager, R., Soumerai, T., Nissan, M.H., et al. (2017). OncoKB: a precision oncology knowledge base. *JCO Precis. Oncol.*, 1–16. <https://doi.org/10.1200/PO.17.00011>.
- Cheng, T.-Y.D., Cramb, S.M., Baade, P.D., Youlten, D.R., Nwogu, C., and Reid, M.E. (2016). The international epidemiology of lung cancer: latest trends, disparities, and tumor characteristics. *J. Thorac. Oncol.* 11, 1653–1671. <https://doi.org/10.1016/j.jtho.2016.05.021>.
- Chollet, F. (2015). Keras. <https://github.com/fchollet/keras>.
- Coudray, N., Ocampo, P.S., Sakellaropoulos, T., Narula, N., Snuderl, M., Fenyö, D., Moreira, A.L., Razavian, N., and Tsirigos, A. (2018). Classification and mutation prediction from non-small cell lung cancer histopathology images using deep learning. *Nat. Med.* 24, 1559. <https://doi.org/10.1038/s41591-018-0177-5>.
- Deng, J., Dong, W., Socher, R., Li, L., Li, K., and Fei-fei, L. (2009). Imagenet: a large-scale hierarchical image database. In 2009 IEEE Conference on computer vision and pattern recognition (IEEE), pp. 248–255.
- Gao, Y., Beijbom, O., Zhang, N., and Darrell, T. (2015). Compact bilinear pooling, Preprint at. ArXiv, 151106062 Cs.
- Hanahan, D., and Weinberg, R.A. (2011). Hallmarks of cancer: the next generation. *Cell* 144, 646–674. <https://doi.org/10.1016/j.cell.2011.02.013>.
- Hatipoglu, N., and Bilgin, G. (2016). Feature extraction for histopathological images using convolutional neural network. In 2016 24th signal processing and communication application Conference (SIU), pp. 645–648. <https://doi.org/10.1109/SIU.2016.7495823>.
- He, K., Fan, H., Wu, Y., Xie, S., and Girshick, R. (2020). Momentum contrast for unsupervised visual representation learning, Preprint at. ArXiv, 191105722 Cs.
- Homer, N., and Nelson, S.F. (2010). Improved variant discovery through local re-alignment of short-read next-generation sequencing data using SRMA. *Genome Biol* 11, R99.
- Huang, G., Liu, Z., van der Maaten, L., and Weinberger, K.Q. (2016). Densely connected convolutional networks, Preprint at. ArXiv, 160806993 Cs.
- Kakiuchi, M., Nishizawa, T., Ueda, H., Gotoh, K., Tanaka, A., Hayashi, A., Yamamoto, S., Tatsuno, K., Katoh, H., Watanabe, Y., et al. (2014). Recurrent gain-of-function mutations of RHOA in diffuse-type gastric carcinoma. *Nat. Genet.* 46, 583–587. <https://doi.org/10.1038/ng.2984>.
- Kather, J.N. (2019). Histological images for MSI vs. MSS classification Gastro-intest. Cancer, FFPE samples. <https://doi.org/10.5281/zenodo.2530835>.
- Kather, J.N., Heijl, L.R., Grabsch, H.I., Loeffler, C., Echle, A., Muti, H.S., Krause, J., Niehues, J.M., Sommer, K.A.J., Bankhead, P., et al. (2020). Pan-cancer image-based detection of clinically actionable genetic alterations. *Nat. Cancer*, 1–11. <https://doi.org/10.1038/s43018-020-0087-6>.
- Kim, S.T., Cristescu, R., Bass, A.J., Kim, K.-M., Odegaard, J.I., Kim, K., Liu, X.Q., Sher, X., Jung, H., Lee, M., et al. (2018). Comprehensive molecular characterization of clinical responses to PD-1 inhibition in metastatic gastric cancer. *Nat. Med.* 24, 1449–1458. <https://doi.org/10.1038/s41591-018-0101-z>.
- Lin, T.-Y., RoyChowdhury, A., and Maji, S. (2015). Bilinear CNNs for fine-grained visual recognition, Preprint at. arXiv, 150407889 Cs.
- Liu, W., Juhas, M., and Zhang, Y. (2020). Fine-Grained breast cancer classification with bilinear convolutional neural networks (BCNNs). *Front. Genet.* 11. <https://doi.org/10.3389/fgene.2020.547327>.
- Maile, E.J., Barnes, I., Finlayson, A.E., Sayeed, S., and Ali, R. (2016). Nervous system and intracranial tumour incidence by ethnicity in England, 2001–2007: a descriptive epidemiological study. *Plos One* 11, e0154347. <https://doi.org/10.1371/journal.pone.0154347>.
- Masaki, T. (2006). Tumor Budding in Colorectal Cancer: Recent Progress in Colorectal Cancer Research (Nova Publishers).
- McInnes, L., Healy, J., and Melville, J. (2018). UMAP: uniform manifold approximation and projection for dimension reduction, Preprint at. arXiv, 180203426 Cs Stat.
- Molin, M.D., Matthaei, H., Wu, J., Blackford, A., Debeljak, M., Rezaee, N., Wolfgang, C.L., Butturini, G., Salvia, R., Bassi, C., et al. (2013). Clinicopathological correlates of activating GNAS mutations in intraductal papillary mucinous neoplasm (IPMN) of the pancreas. *Ann. Surg. Oncol.* 20, 3802–3808. <https://doi.org/10.1245/s10434-013-3096-1>.
- Muja, M., and Lowe, D.G. (2009). Fast approximate nearest neighbors with automatic algorithm configuration. In In VISAPP International Conference on Computer Vision Theory and Applications, pp. 331–340.
- Nagpal, K., Foote, D., Liu, Y., Po-Hsuan, Chen, Wulczyn, E., Tan, F., Olson, N., Smith, J.L., Mohtashamian, A., Wren, J.H., et al. (2018). Development and validation of a deep learning algorithm for improving Gleason scoring of prostate cancer.
- Paszke, A., Gross, S., Chintala, S., Chanan, G., Yang, E., DeVito, Z., Lin, Z., Desmaison, A., Antiga, L., and Lerer, A. (2017). Automatic differentiation in PyTorch. <https://openreview.net/forum?id=BJJsrnfCZ>.
- Pedregosa, F., Varoquaux, G., Gramfort, A., Michel, V., Thirion, B., Grisel, O., Blondel, M., Prettenhofer, P., Weiss, R., Dubourg, V., et al. (2011). Scikit-learn: machine learning in Python. *J. Mach. Learn. Res.* 12, 2825–2830.

- Pharoah, P.D.P., Guilford, P., and Caldas, C. (2001). Incidence of gastric cancer and breast cancer in CDH1 (E-cadherin) mutation carriers from hereditary diffuse gastric cancer families. *Gastroenterology* 121, 1348–1353. <https://doi.org/10.1053/gast.2001.29611>.
- Phillips, S.M., Banerjee, A., Feakins, R., Li, S.R., Bustin, S.A., and Dorudi, S. (2004). Tumour-infiltrating lymphocytes in colorectal cancer with microsatellite instability are activated and cytotoxic. *Br. J. Surg.* 91, 469–475. <https://doi.org/10.1002/bjs.4472>.
- Reinhard, E., Adhikmin, M., Gooch, B., and Shirley, P. (2001). Color transfer between images. *IEEE Comput. Graph. Appl.* 21, 34–41. <https://doi.org/10.1109/38.946629>.
- Riaz, N., Bleuca, P., Lim, R.S., Shen, R., Higginson, D.S., Weinhold, N., Norton, L., Weigelt, B., Powell, S.N., and Reis-Filho, J.S. (2017). Pan-cancer analysis of bi-allelic alterations in homologous recombination DNA repair genes. *Nat. Commun.* 8, 1–7. <https://doi.org/10.1038/s41467-017-00921-w>.
- Sandler, M., Howard, A., Zhu, M., Zhmoginov, A., and Chen, L.-C. (2018). MobileNetV2: inverted residuals and linear bottlenecks. Preprint at. ArXiv, 180104381 Cs.
- Schaumberg, A.J., Rubin, M.A., and Fuchs, T.J. (2017). H&E-stained whole slide image deep learning predicts SPOP mutation state in prostate cancer. Preprint at. bioRxiv, 064279. <https://doi.org/10.1101/064279>.
- Simonyan, K., and Zisserman, A. (2014). Very deep convolutional networks for large-scale image recognition. Preprint at. ArXiv, 14091556.
- Stringer, C., Wang, T., Michaelos, M., and Pachitariu, M. (2021). Cellpose: a generalist algorithm for cellular segmentation. *Nat. Methods* 18, 100–106. <https://doi.org/10.1038/s41592-020-01018-x>.
- Szegedy, C., Vanhoucke, V., Ioffe, S., Shlens, J., and Wojna, Z. (2015). Rethinking the inception architecture for computer vision. Preprint at. ArXiv, 151200567 Cs.
- The Cancer Genome Atlas Research Network; Bass, A.J., Thorsson, V., Shmulevich, I., Reynolds, S.M., Miller, M., Bernard, B., Hinoue, T., Laird, P.W., Curtis, C., et al. (2014). Comprehensive molecular characterization of gastric adenocarcinoma. *Nature* 513, 202–209. <https://doi.org/10.1038/nature13480>.
- Wang, C., Shi, J., Zhang, Q., and Ying, S. (2017). Histopathological image classification with bilinear convolutional neural networks. *Annu. Int. Conf. IEEE Eng. Med. Biol. Soc.* 2017, 4050–4053. <https://doi.org/10.1109/EMBC.2017.8037745>.
- Weinstein, J.N., Collisson, E.A., Mills, G.B., Shaw, K.R.M., Ozenberger, B.A., Ellrott, K., Shmulevich, I., Sander, C., and Stuart, J.M. (2013). The cancer genome atlas pan-cancer analysis project. *Nat. Genet.* 45, 1113–1120.
- Yosinski, J., Clune, J., Nguyen, A., Fuchs, T., and Lipson, H. (2015). Understanding neural networks through deep visualization. Preprint at. ArXiv, 150606579 Cs.
- Youden, W.J. (1950). Index for rating diagnostic tests. *Cancer* 3, 32–35. [https://doi.org/10.1002/1097-0142\(1950\)3:1<32::AID-CNCR2820030106>3.CO;2-3](https://doi.org/10.1002/1097-0142(1950)3:1<32::AID-CNCR2820030106>3.CO;2-3).
- Zanet, J., Pibre, S., Jacquet, C., Ramirez, A., de Alborán, I.M., and Gandarillas, A. (2005). Endogenous Myc controls mammalian epidermal cell size, hyperproliferation, endoreplication and stem cell amplification. *J. Cell Sci.* 118, 1693–1704. <https://doi.org/10.1242/jcs.02298>.
- Zoph, B., Vasudevan, V., Shlens, J., Le, Q.V., 2017. Learning transferable architectures for scalable image recognition.

STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
anti-pan-cytokeratin	DAKO	monoclonal, clone AE1/AE3, IS05330-2J
anti-PD-L1	Roche Diagnostics	monoclonal, #790-4905; RRID:AB_2819099
anti-CD8	DAKO	monoclonal, clone CB/144B, M710301-2
Biological samples		
Human gastric cancer biospecimens	This paper	The University of Tokyo Hospital: Collected under G3521-(28)
Human cancer tissue microarrays	This paper	The University of Tokyo Hospital: Collected under #2381-(8)
Deposited data		
Histology images from uniform tumor regions in TCGA Whole Slide Images	This paper	doi:10.5281/zenodo.5889558
TCGA pan-cancer whole slide images	NCI GDC	https://portal.gdc.cancer.gov/legacy-archive/
TCGA pan-cancer RNA-seq data	PanCanAtlas	RNA (Final) https://gdc.cancer.gov/about-data/publications/pancanatlas
Gleason 2019	University of British Columbia	https://gleason2019.grand-challenge.org/
TCGA pan-cancer histological subtypes	PanCanAtlas	Clinical with Follow-up https://gdc.cancer.gov/about-data/publications/pancanatlas
OncoKB	the Memorial Sloan Kettering Cancer Center	https://www.oncokb.org/
TCGA Fusion Genes	the US Jackson Laboratory	http://www.tumorfusions.org/
TCGA mutational signatures	PCAWG consortium	PCAWG MUTATIONAL SIGNATURES https://dcc.icgc.org/releases/PCAWG
Contribution of each mutational signature and large-scale state transitions (LST) information	the Memorial Sloan Kettering Cancer Center	https://github.com/riazn/biallelic_hr
MSI/MSS label in TCGA STAD and COAD/READ the mutation prediction performance of ShuffleNet-based classifier	University Hospital RWTH Aachen (Kather et al., 2020)	https://doi.org/10.5281/zenodo.2530835 Source Data Extended Data Figure 1 (https://static-content.springer.com/esm/art%3A10.1038%2Fs43018-020-0087-6/MediaObjects/43018_2020_87_MOESM7_ESM.xlsx)
world population data	The World Bank	https://data.worldbank.org/
The numbers of cancer incidences	World Health Organization (WHO)	https://gco.iarc.fr/today/online-analysis-map
Software and algorithms		
keras-OpenSlideGenerator	This paper	https://github.com/quolc/keras-OpenSlideGenerator
Deep texture representation	This paper	doi:10.5281/zenodo.5889561
Keras (version 2.2.4)	(Chollet, 2015)	https://github.com/keras-team/keras
Tensorflow (version 1.10.0)	(Abadi et al., 2016)	https://arxiv.org/abs/1603.04467
Pytorch Lightning (version 1.2.10)	The PyTorch Lightning team	https://www.pytorchlightning.ai/
StainTools (version 2.1.2)	Peter Byfield	https://github.com/Peter554/StainTools
Python UMAP library	(McInnes et al., 2018)	https://github.com/lmcinnes/umap
scikit-learn (version 0.20.0)	(Pedregosa et al., 2011)	https://github.com/scikit-learn/scikit-learn
Cellpose (version 0.6.5)	(Stringer et al., 2021)	https://github.com/MouseLand/cellpose
R (version 3.6.3)	R Development Core Team	https://www.r-project.org/
Cgdsr	the Memorial Sloan Kettering Cancer Center	https://github.com/cBioPortal/cgdsr

(Continued on next page)

Continued		
REAGENT or RESOURCE	SOURCE	IDENTIFIER
oncokb-annotator	the Memorial Sloan Kettering Cancer Center	https://github.com/oncokb/oncokb-annotator
gradcam_plus_plus-pytorch	WonKwang Lee	https://github.com/1Konny/gradcam_plus_plus-pytorch
pytorch (version 1.0)	(Paszke et al., 2017)	https://github.com/pytorch/pytorch
Pyflann	Gefu Tang	https://github.com/primetang/pyflann
Other		
Luigi	This paper	https://luigi-pathology.com/

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Shumpei Ishikawa (ishum-prm@m.u-tokyo.ac.jp).

Materials availability

This study did not generate new unique reagents.

Data and code availability

- The histology images from uniform tumor regions in TCGA Whole Slide Images have been deposited at Zenodo and is publicly available as the date of publication. DOIs are listed in the key resources table.
- All original code has been deposited at Zenodo and is publicly available as of the date of publication. DOIs are listed in the key resources table.
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

EXPERIMENTAL MODEL AND SUBJECT DETAILS

Unsupervised DTR analysis for gastric cancer specimens treated with anti-PD-1 checkpoint inhibitor

We retrospectively analyzed prospectively collected data from 55 consecutive patients with recurrent or metastatic gastric cancer who began treatment with nivolumab at the University of Tokyo Hospital between October 2017 and March 2020. From 35 out of the 55 patients, formalin fixed paraffin embedded (FFPE) tissues by gastric tissue biopsy between June 2010 and June 2018 were retrospectively collected (n = 9 female and n = 26 male, mean age 66.8 [SD = 12.8], [Table S2](#)). As all the biopsy samples analyzed in this study were obtained at the time of initial diagnosis of gastric cancer, the patients did not receive any gastric surgery, chemotherapy, or radiotherapy at the time of the biopsy.

Nivolumab was administered at a dose of 3 mg/kg body weight every 2 weeks from October 2017 to mid-September 2018 and then administered at a fixed dose of 240 mg every 2 weeks from mid-September 2018 to December 2020 (data cutoff for the present study), according to the change in guidelines approved by the Japanese MHLW. Nivolumab treatment continued until unacceptable adverse effects occurred or the disease progressed. Nivolumab was continued beyond disease progression if the physician considered that it was providing clinical benefit. Informed consent was obtained from each subject, and the study was approved by the Institutional Review Board of the University of Tokyo Hospital.

External validation cohort of genomic aberration prediction

Fresh frozen gastric cancer and paired normal stomach tissues were obtained from patients who underwent gastrectomy at the University of Tokyo Hospital (n = 17 female and n = 29 male, mean age 67.8 [SD = 11.9]). Informed consent was obtained from each subject, and this study was approved by the institutional review boards at the University of Tokyo. Genomic DNA was extracted using QIAamp DNA Mini kit (Qiagen, Hilden, Germany) according to the manufacturer's instructions.

METHOD DETAILS

TCGA whole slide image dataset

WSIs of TCGA dataset representing 32 solid cancer types were downloaded from Genomics Data Commons legacy database from December 1, 2016 to June 19, 2017. In total, 9,662 diagnostic slides (with filename containing “DXn,” where n represents the slide number) from 7,951 patients in SVS format were then processed for annotation. The number of slides for each cancer type is shown in Table S1.

Image preprocessing of TCGA images

For each slide, at least three representative tumor regions were selected as polygons by two trained pathologists using web browser-based software developed for this purpose. To select regions suitable for DTRs, uniform tumor regions were selected while avoiding as much as possible any regions with noncancerous structures. Trained pathologists excluded 926 slides with the following features before analysis: (1) poor staining, (2) not a WSI but rather a captured image with an unknown magnification level, (3) being out of focus, (4) lacking cancerous regions, and (5) consisting of incorrect cancer types. Notably, features 1–4 are unrelated to the tumor’s cellularity and clinical or molecular features. Feature 5 should be removed because it causes incorrect interpretation. Finally, 8,736 diagnostic slides from 7,175 patients were used for downstream analysis.

For DTR visualization, 10 patches from each of the annotated regions were arbitrarily cropped with random angles at 6 magnification levels (from 128×128 to $256 \times 256 \mu\text{m}$) using keras-OpenSlideGenerator (<https://github.com/quolc/keras-OpenSlideGenerator>). The pixel size corresponding to $S \mu\text{m}$ was calculated as S/m , where m is micron per pixel (mpp), ranging from 0.11504 to 0.504 with a median value of 0.252. The mpp values were included as metadata (“aperio.MPP” or “openslide.mpp-x”) in the SVS files. Images with regions too small to crop images were skipped. Forty-two WSIs lacking mpp information were removed. Each patch was selected so as not to include areas outside the annotated region; selected regions were resized to 256×256 pixels. Consequently, the number of patches subjected to further analysis ranged from 264,110 to 271,700.

For the CBIR application, the number of patches per region was limited to three to reduce memory consumption and search time for the system. As a result, 475,701 total patches were included in the database.

For the genome prediction tasks, the same dataset used within DTR visualization was utilized, but adenocarcinoma cases of CESC and ESCA were removed to avoid classification bias due to major histological category (e.g. adenocarcinoma and squamous cell carcinoma). The resulting number of patches ranged from 261,750 to 269,250.

Deep texture representations (DTRs)

DTRs were represented by the correlation matrix $G \in \mathbb{R}^{N \times N}$ of the filter responses in any layer of the convolutional neural network (CNN), where N is the total number of filters in the layer. The (i, j) -th entry G_{ij} is the inner product between the i -th and j -th vectorized feature maps in the layer:

$$G_{ij} = \sum_k F_{ik} F_{jk}$$

where $F_{ik} \in \mathbb{R}$, represents the k -th component in the vectorized feature maps of i -th filter. This manipulation—also known as bilinear pooling—produces spatially-orderless image representations. This process results in the discarding of some spatial information, and thus the representations contain spatial invariance, which appears to be suitable for cancer histology images as the similarity between two such images should not be affected by the relative location of cellular and extracellular components. In addition, as the representations are too high dimensional for practical application, CBP using Random Maclaurin approximation (Gao et al., 2015, p. 20) was adopted to estimate the original high dimensional representations in lower final dimensions. CBP is an approximation bilinear pooling method utilizing two random matrices, and unlike other dimensionality reduction methods, such as principal component analysis, it does not require pretraining. Prior to the final representation, we added an element-wise signed square root layer and an instance-wise normalization as previously shown elsewhere (Lin et al., 2015).

Selection of optimal neural network architecture and layer

DTRs can be computed on any convolutional layers in any CNN structure. Due to the lack of CNNs pretrained with a large number of histopathology images, we have employed a model trained with an ImageNet dataset (Deng et al., 2009), which comprises over 10 million images representing 1,000 classes and exhibits satisfactory performance within histopathology image analysis. To select an optimal network architecture and a corresponding layer that allows DTRs to capture characteristics of histology images properly, we have evaluated the accuracy of DTRs from various representative pretrained models to capture the similarity between histopathology images (Figure S1A).

First, 21 images for each of the 32 cancer types were randomly selected from the preprocessed 271,700 patches ($128 \times 128 \mu\text{m}$), of which one image per cancer type was chosen as a query. From the remaining 640 images after removing the 32 images used as the query, three trained pathologists independently selected three images that were histologically similar to each query and ranked them according to the similarity to the query image.

We used DTRs, implemented using Keras version 2.2.4 with Tensorflow 1.10.0 backend, for each of the representative convolutional layers of six common CNN architectures VGG16 (Simonyan and Zisserman, 2014), MobileNet V2 (Sandler et al., 2018), Inception V3 (Szegedy et al., 2015), NasNet-A Large (Zoph et al., 2017), DenseNet201 (Huang et al., 2016), and Inception ResNet V2 (Szegedy et al., 2015). First, cosine similarities between the q^{th} query image or the one rotated by 90° and the selected i^{th} image were calculated, and the larger value was retained as $s_{q,i}$.

$$s_{q,i} = \max(\text{sim}(Q_{q,0}, I_i), \text{sim}(Q_{q,90}, I_i)),$$

where $Q_{q,0}$ is a q^{th} query image, $Q_{q,90}$ is a q^{th} query image rotated 90° , I is an evaluated image, and sim is a cosine similarity function.

Here, we denote a vector composed of $s_{q,i}$ by S_q .

$$S_q = (s_{q,1}, s_{q,2}, s_{q,3})$$

Then the vector of Spearman rank correlation coefficient with the rank by a trained pathologist was calculated as follows:

$$R_{p,q} = (r_p(Q_{q,0}, I_1), r_p(Q_{q,0}, I_2), r_p(Q_{q,0}, I_3)),$$

where r_p is a rank assigned by the p^{th} pathologist. Finally, the mean correlation over the three pathologists and query images were used as the evaluation index;

$$\frac{1}{N_q N_p} \sum_{q=1}^{N_q} \sum_{p=1}^{N_p} \text{corr}(S_q, R_{p,q}),$$

where N_q is the total number of query images (=32) and N_p is the total number of trained pathologists (=3).

To evaluate the robustness of the optimization, we replaced each image patch in the queries or the selected patches with another randomly selected from the same region, and then we used Spearman rank correlation and the same rank by the pathologists to conduct statistical assessments. This entire process was repeated 1,000 times.

We also compared the performance of DTRs with two deep neural network-based features (from a supervised or self-supervised pretrained CNN models fine-tuned with histopathological images).

To get the deep features, we fine-tuned VGG16 network architectures with an input size of 256×256 , which were pretrained on ImageNet. Due to the limited number of training samples, only the weights in the last 6 convolutional layers were trainable whereas all other weights were frozen. Of the previously described 271,700 patches ($128 \times 128 \mu\text{m}$) representing 32 cancer types from 7175 patients, the patches from 75% of the patients were used for training and the patches from the remaining 25% of the patients were used for validation. To reduce the effect of class imbalance, the number of patches per class was equalized by oversampling. We used data augmentation to virtually increase the size of the training data. It includes random vertical or horizontal flip with additional random hue and saturation offsets. To avoid overfitting, we used categorical cross-entropy loss with label smoothing with the amount of smoothing of 0.05. We trained the network using the Adam optimization method with a learning rate of (0.1, 0.01, 0.001, 0.0001) and a batch size of 96. The training was run for 5 epochs to optimize only the last fully connected layers and then run for 25 epochs with 4 epoch warm start to optimize the rest trainable layers. The strategy was tested for the cancer type classification, and the models with the highest top-1 validation accuracy (30.7% for VGG16) were chosen as the final models. The outputs of the global average pooling and CBP after the convolution layers were used as deep features.

Momentum Contrast (MoCo) (He et al., 2020) is a state-of-the-art method for unsupervised visual representation learning. Here, we used MoCo v2 implemented in Pytorch Lightning version 1.2.10. Of the previously described 271,700 patches ($128 \times 128 \mu\text{m}$) representing 32 cancer types from 7,175 patients, the patches from 75% to 25% of the patients were used for training and validation, respectively. The VGG16 model was trained from scratch using a stochastic gradient descent optimizer with a learning rate of 0.331, which was optimized by Pytorch Lightning's built-in LR finder, and a batch size of 256. The training was ended at the 200th epoch and the model with the minimum validation loss was used for further analysis.

Evaluation of the robustness of DTRs to image perturbation

We evaluated the robustness of DTRs to various image perturbations, including brightness, contrast, sharpness, and simulated color variation. First, we applied image enhancement of brightness, contrast, or sharpness using ImageEnhance Module of Pillow package version 8.0.1 with an enhance factor from -0.5 to 0.5 in increments of 0.1 (except for 0 , i.e., unchanged). Otherwise, we applied stain augmentation, using the staintools package version 2.1.2 and the method "vahadane," to each of 500 randomly selected patches from TCGA dataset. Stain augmentation was applied 100 times. The cosine similarities of DTRs were then evaluated in a comparison of the original image and the perturbed images. Patches randomly selected from the same region as that of the original image were used for comparison.

UMAP visualization of DTRs

the nonlinear dimension reduction algorithm uniform manifold approximation and projection (UMAP) (McInnes et al., 2018) was used to visualize DTRs. The averaged DTRs in each case or region were used so that one patient or region would occupy one point in the 2D space. Python's UMAP library (<https://github.com/lmcinnes/umap>) was used for calculation with $n_neighbors = 10$, $min_dist = 0.2$

and metric-“cosine distance” for the visualization of a pan-cancer dataset. We used different parameters for the visualization of specific cancer types, as the data size varied for data of different cancer types, for example: $n_neighbors = 13$, and $min_dist = 0.8$ for prostate adenocarcinoma (PRAD) and STAD; and $n_neighbors = 13$ and $min_dist = 0.4$ for THCA. The Gleason score for PRAD and standard Lauren classification for STAD in each region were evaluated by trained pathologists.

Prediction of gleason score for PRAD and lauren classification for STAD

For the Gleason score (g3, g4, or g5) in PRAD and standard Lauren classification (intestinal or diffuse) in STAD, supervised classification models using logistic regression, where the average value of 1,024-dimensional DTRs from each region was used as features, were developed and the performance was evaluated in terms of AUC. Two-fold cross validation for the cases—and not for regions, as to avoid regions from the same case being contained in both training and test data—was performed for each classification model using the GroupShuffledSplit function implemented in scikit-learn. The prediction was made for a magnification level of 256 μm . For multiclass prediction for Gleason score, one versus left classifiers were developed for each Gleason score, and AUC for macro-average ROC was used for evaluation. Finally, the mean AUC for the two cross-validations was reported. Similarly, DTRs from the color-normalized images were also evaluated for each classification with exactly the same settings, including the training and test dataset cases. Python version 3.6.3 and the machine-learning library scikit-learn version 0.20.0 were used for logistic regression, cross validation, and AUC calculation. For logistic regression, the LogisticRegression function with penalty = “l2,” was applied.

Cell count estimate using Cellpose

To show that DTRs capture complex morphology rather than just cell count per patch, we evaluated the cellularity of image patches. Specifically, we compared the median cell count per patch among different categories for Gleason grading in prostate adenocarcinoma and for Lauren classification in stomach adenocarcinoma from TCGA dataset. We applied Cellpose version 0.6.5 (Stringer et al., 2021), a robust state-of-the-art cell-detection algorithm, to the image patches. We chose a model with the following parameters: “diameter = 8, channels = [3,0], batch_size = 64, and cellprob_threshold = 0.1”; this model detected the nucleus with acceptable accuracy after some trials. We also created the classifier for Gleason grading and Lauren classification using median cell count as a feature. The machine learning algorithm and the training samples were the same as those presented in Figure 3A.

DTR distribution for the independent external validation cohort for prostate cancer

We analyzed the Gleason 2019 dataset (Nagpal et al., 2018) as an external validation cohort for prostate cancer. For each core of tissue microarrays from the annotated training dataset, the Gleason grade that occupied the largest area in the regions annotated by six pathologists was chosen as the Gleason grade for the core. The image patches of 1,024 $\times$ 1,024 pixels corresponding to 256 $\times$ 256 μm were then extracted with 512-pixel stride if more than 90% of the patch was annotated with the determined Gleason grade by the majority of pathologists. Finally, DTRs were calculated for the image patch, and the mean DTRs were plotted using UMAP with $n_neighbors = 15$, $min_dist = 0.3$, and metric-“cosine distance” for the visualization.

Gene expression analysis to identify rare subtypes of pancreas adenocarcinoma

Pan-cancer gene expression profile was downloaded from National Cancer Institute (NCI) Genomic Data Commons (<https://gdc.cancer.gov/about-data/publications/pancanatlas>) on November 12, 2018 and Z score transformation for each gene was performed across PAAD cases. The following genes and the cutoff values were used to identify subtypes of PAAD cases.

- Intestinal differentiation: normalized CDX2 expression level >4
- Adenosquamous differentiation: mean normalized expression of three squamous cell markers (KRT5, KRT17, TP63) >4 .

Additionally, cases with Zscored_SCN_score >2 in the previous study (Balanis et al., 2019) were considered as neuroendocrine tumor.

Unsupervised DTR analysis for gastric cancer specimens treated with anti-PD-1 checkpoint inhibitor

WSIs from H&E-stained sections of the FFPE tissues were analyzed to investigate the correlation of their DTRs with the response to PD-1 inhibitor nivolumab. Slides were digitized using a Hamamatsu Nanozoomer 2.0 HT whole slide scanner. For each of the resulting WSIs in Hamamatsu ndpi format, at least one representative cancerous region was selected as polygons using NDPview2 viewing software (Hamamatsu Photonics) and Lauren classification (intestinal, indeterminate, and diffuse) was evaluated for the regions by trained pathologists. Then, 30 patches sized 128 $\times$ 128 μm were arbitrarily cropped using random angles from each of the annotated region using keras-OpenSlideGenerator and were then resized to 256 $\times$ 256 pixels and the DTRs were calculated. OS and PFS were analyzed with the use of a two-sided log rank test stratified according to the value of second principal component of DTRs. Survival curves and rates were estimated with the use of the Kaplan–Meier method. Data were censored on December 31, 2020. Patients who were lost to follow-up were censored at the date of last contact or follow-up. OS was calculated from the date of nivolumab initiation to the date of death from any cause. Patients who were alive on December 31, 2020 were censored for OS analysis. Tumor response was evaluated according to the Response Evaluation Criteria in Solid Tumors, version 1.1, based on computed tomographic (CT) findings. The best overall response was assessed as complete response (CR), partial response (PR), stable disease (SD) or progres-

sive disease (PD). Patients whose diseases clinically progressed could be diagnosed as PD without CT finding. Correlation of Lauren classification (indeterminate vs. intestinal or diffuse) or PC2 (the cutoff value was determined so that the number of positive samples is the same as that of indeterminate samples) to the response to nivolumab treatment (PD vs. CR, PR or SD) was evaluated by two-sided Fisher's exact test and phi correlation coefficient with standard and "psych" in R library, respectively.

Immunohistochemistry

Tissue specimens were obtained by gastric biopsies in the University of Tokyo Hospital, fixed by 10% formalin, and embedded in paraffin. The paraffin blocks were then sliced into 4 μ m thick slides, and subsequently dewaxed and rehydrated using xylene and graded ethanol, respectively, followed by washing with distilled water. Antigen retrieval was performed by autoclave for 5 min at 120°C immersed in citrate buffer (pH 6.0) (Abcam, Cambridge, UK). Endogenous peroxidase activity was blocked with 0.3% hydrogen peroxide (Sigma-Aldrich, MO, USA) for 15 min followed by washing with distilled water. Nonspecific protein-protein reaction was blocked by incubating the slides in 2% bovine serum albumin (Sigma-Aldrich)/Phosphate Buffered Saline for 15 min at room temperature. The following primary antibodies were used and were applied overnight at 4°C: monoclonal mouse IgG anti-pan-cytokeratin, clone AE1/AE3 (without dilution; IS05330-2J; DAKO, Santa Clara, CA, USA); Monoclonal rabbit anti-PD-L1 antibody (without dilution; #790-4905; Roche Diagnostics, Pleasanton, CA, USA); and monoclonal mouse IgG anti-CD8, clone CB/144B (1: 200 dilution; M710301-2; DAKO, Santa Clara, CA, USA). HistostarTM Immunostaining Reagent (#8460; Medical and Biological Laboratories co., Ltd., Aichi, Japan) and HistostarTM DAB substrate solution (#8469; Medical and Biological Laboratories co., Ltd.) were used for detecting immunohistochemistry signals. Nuclear staining was performed with hematoxylin solution (Sakura finetek Japan, Tokyo, Japan).

Staining intensity was evaluated by two pathologists. Infiltration levels of CD8+ T cells in the tumor microenvironments were evaluated and graded into 3 categories 1, 2, and 3 corresponding to mild, moderate, and severe infiltration statuses, respectively. PD-L1 staining of the cellular membrane of tumor cells were graded into 4 categories, i.e., 0 (none), 1 (weakly positive), 2 (moderately positive), and 3 (highly positive), respectively. Correlation of the CD8 or PD-L1 grade to the response to nivolumab treatment (PD vs. CR, PR or SD) was evaluated by two-sided Fisher's exact test and phi correlation coefficient with standard and "psych" in R library, respectively.

Color normalization

Color normalization was performed using the method of (Reinhard et al., 2001) implemented in StainTools (<https://github.com/Peter554/StainTools>). Reference patches were selected from cancerous regions in TCGA dataset by a pathologist (TCGA-3M-AB46-01Z-00-DX1.svs for STAD, TCGA-2A-A8VL-01Z-00-DX1.svs for PAAD, and TCGA-2J-AAB6-01Z-00-DX1.svs for PRAD), and color normalizations were performed separately for each cancer type. It should be noted that the color normalization was performed only in the part of Figure 2C and not in the rest of the paper.

Independent validation cohort for CBIR evaluation

We analyzed WSIs from patients with various cancers (glioma, kidney renal clear cell carcinoma, lung adenocarcinoma, breast adenocarcinoma, colon adenocarcinoma, head and neck squamous cell carcinoma, prostate adenocarcinoma, sarcoma, and testicular germ cell tumor) diagnosed at the University of Tokyo Hospital after 1955. Because the data were reported anonymously, the requirement for informed consent was waived. The study was approved by the Institutional Review Board of the University of Tokyo.

Slides of H&E-stained tissue microarrays from the FFPE tissues were digitized using a Hamamatsu Nanosizer S60 whole slide scanner. For each of the resulting WSIs in Hamamatsu ndpi format, a representative cancerous region for each cancer type was captured as a 256 $\times$ 256 μ m square with NDPview2 viewing software (Hamamatsu Photonics). The image patches, resized to 256 $\times$ 256 pixels, were converted to DTRs and input into the CBIR system. The cases used for the DTR model selection were excluded from the retrieved results.

Genomic aberration and clinical data of TCGA

Somatic mutations

For cancer driver genes, actionable alterations and all alterations (driver alterations) were downloaded from the OncoKB database (Chakravarty et al., 2017) of the Memorial Sloan Kettering Cancer Center. There were 44 and 322 genes found for actionable and driver alterations, respectively. SNVs, indels, CNVs and gene fusions were used for analyses. TCGA cases with each driver mutation were retrieved from "tcga pan can atlas 2018" database using the "cgdsr" Bioconductor package. The modified script of "AnnotatorCore.py" of oncoKB-annotator (<https://github.com/oncoKB/oncoKB-annotator>) was used to classify the mutations into actionable or driver alterations, as well as others. TCGA cases with fusion genes were downloaded from Tumor Fusion Gene Data Portal (<https://tumorfusions.org/>) developed by the US Jackson Laboratory. Driver genes mutated in at least three patients for at least one of the cancer types were used for evaluation, which resulted in 208 annotated and 30 actionable genes.

Mutational signatures

Single base substitution (SBS) mutational signatures identified by SigProfiler in a previous study (Alexandrov et al., 2020) were analyzed. The SBS was seen in at least 5% of WGS samples in the cancer type in PCAWG signature studies (Alexandrov et al., 2020). Because the cancer types used in the TCGA and PCAWG did not always match, the non-apparent correspondence between the TCGA and PCAWG cancer types was determined as follows: BRCA = Breast-AdenoCA and Breast LobularCA and SARC = Soft-

Tissue-Liposarcoma, TGCT, THYM, UVM, KIRP, ACC, MESO, ESCA (esophageal adenocarcinoma was included, but only esophageal squamous cell carcinoma cases were used in our analysis) and PCPG were excluded in this analysis due to the lack of WGS samples in the PCAWG study. Then, several SBS signatures with similar etiologies were merged (Table S9). For each SBS signature, positive cases were defined as having at least ten mutations derived from the signature. Only those signatures that were found to be positive or negative in more than two patients in the cancer type were used for the evaluation, which resulted in 13 SBS signatures.

BRCAness genomic signature

Based on a previous paper (Riaz et al., 2017), we defined BRCAness as being positive when a case had the largest number of signature 3 mutations among all signatures, and the number of large-scale state transitions (LST) was ≥ 15 . Contribution of each mutational signature and LST information was downloaded from https://github.com/riazn/biallelic_hr.

Microsatellite instability

Definition of cases with MSI or defined as being microsatellite stable (MSS) for STAD and COAD/READ were taken from prior research (Kather, 2019). Briefly, patients who were previously defined as MSI-High (MSI-H) or patients with a mutation count of $>1,000$ were defined as having MSI. The MSI/MSS label for each patient was downloaded from the corresponding Zenodo repository (Kather, 2019).

Histological subtype

Histological subtypes of the TCGA cases were downloaded from National Cancer Institute (NCI) Genomic Data Commons (<https://gdc.cancer.gov/about-data/publications/pancanatlas>) on November 12, 2018. The term "histological_type" in clinical_PANCAN_patient_with_followup.tsv was used for query.

Prediction of genomic features from histopathology images using DTRs and supervised machine learning

The presence of a driver gene mutation or mutation signature of each cancer type was predicted using logistic regression, where 1,024-dimensional DTRs from histopathology images were used as features. Stratified 3-fold cross validation for the cases—and not for images, as to avoid patches from the same case being contained in both training and test data—was performed for each type of mutation and cancer. During the training phase, all the patches were treated independently irrespective of the derived cases, but the prediction as well as AUC score calculation were performed for each case by averaging the probability score of the logistic regression model for all the patches from the case. The prediction was made for six magnification levels (128–256 μm), and the optimal model was selected. Finally, for the genomic aberrations with mean AUC >0.65 , the logistic regression models were trained using all the available data at three magnification levels (128, 192, and 256 μm) and implemented into the web system.

The programming language Python version 3.6.3 and the machine-learning library scikit-learn version 0.20.0 were used for logistic regression, cross validation, and AUC calculation. For logistic regression, SGD Classifier with loss = "log," penalty = "l2," alpha = 0.0001 was applied.

Performance comparison of driver mutation prediction with ShuffleNet-based classifier

We conducted rough comparison of the mutation prediction performance of DTR-based logistic regression classifier with that of ShuffleNet-based classifier previously presented¹⁸, which dealt with the same FFPE diagnostic slides in TCGA dataset. Source Data Extended Data Figure 1 was downloaded and mean AUC of driver mutation prediction (tagged as DRIVER) were compared with that of DTR-based logistic regression using SNV/Indel. CESC cases were removed for evaluation since we did not analyze adenocarcinoma cases while the previous research did. Note that it is not a strict comparison of the method since the selected samples, patches or scales can be different.

External validation cohort of genomic aberration prediction

Whole-exome sequencing and mutation analysis

Whole-exome sequencing was performed for 46 gastric cancers and paired normal gastric tissues as previously described (Kakiuchi et al., 2014). Each DNA sample (1.1 μg) was sheared using a Covaris SS Ultrasonicator (Covaris, MA, USA) as per the manufacturer's instructions. A Sciclone NGS workstation (Caliper Life Sciences, MA, USA) was used to automatically construct the DNA sequence libraries according to the manufacturer's instruction. Exome regions were captured by an Agilent SureSelect Human All Exon V4 Kit and V5+ LincRNA (Agilent Technologies, CA, USA). Sequencing was performed on an Illumina HiSeq 2000 instrument (Illumina, CA, USA) and the provided protocol for producing 2 $\times$ 100-bp paired-end reads. Image analyses and base calling were performed using the Illumina pipeline with default settings.

The reads were aligned to the human reference genome GRCh37/hg19 using the Burrows-Wheeler Aligner and NovoAlign software (Novocraft Technologies, Selangor, Malaysia). After removal of PCR duplicates, the Short-Read Micro re-Aligner method was used to improve variant discovery via local realignments (Homer and Nelson, 2010). Variant calling was performed using an integrated genotyping software (karkinos: <http://github.com/genome-rcast/karkinos>) with default settings as previously reported (Kakiuchi et al., 2014).

Prediction of somatic mutations from histopathology images using DTRs

H&E-stained WSIs from 46 gastric cancer tissues with somatic mutation information were used to validate the classification model for gene mutation prediction from histopathology images using DTRs. Slides were digitized using a Hamamatsu Nanozoomer S60 whole slide scanner. For each of the resulting WSIs in Hamamatsu ndpi format, at least three representative cancerous regions were

selected as polygons by a trained pathologist as performed within the TCGA dataset using the same criteria. Then, 10 patches sized $128 \times 128 \mu\text{m}$ were arbitrarily cropped using random angles from each of the annotated region using keras-OpenSlideGenerator, and were then resized to 256×256 pixels and input to the mutation predictors described above. TP53, CDH1, and PTEN genes were mutated in > two patients and thus were chosen for validation. AUC score was calculated for each case by averaging the probability score of the logistic regression model for all the patches from the same case.

Visualization using Grad-CAM

The Grad-CAM algorithm implemented in pytorch (https://github.com/1Konny/gradcam_plus_plus-pytorch) was used to visualize the relevant regions in histology images for the prediction of genomic aberrations. Because Grad-CAM requires fine-tuned CNN models for accurate visualization, we fine-tuned the VGG16 network pretrained on ImageNet for predicting TP53 mutations in BRCA, MSI status in STAD, IDH1 mutations in GBM and U2AF1 mutations in LUAD, all of which are predictable by the logistic regression model using DTRs. All cases were randomly split into training, validation, and test datasets with a 4:1:1 ratio. The network was trained using Stochastic Gradient Descent optimizer with learning rate = 0.00005 and momentum = 0.99. In the first three epochs, only the fully connected layers were trained, and in the remaining 12 epochs, all layers were trained. The model with minimum validation loss was used for Grad-CAM. Pytorch 1.0 was used to train the network.

Luigi system for CBIR and genomic aberration prediction

Implementation

The Luigi system was developed with a Python backend and Clojure frontend; thus, it can be accessed by any modern web browser on Mac OS X or Windows operating system. Users can search images relevant to a query image from the database, which is comprised of 475,701 patches representing 32 cancer types from TCGA according to the aforementioned selection procedure using intuitive operations. Links for the cBioPortal (Cerami et al., 2012) website are also provided, enabling the users to access clinical, pathological, and molecular information relevant to the retrieved case. The Luigi system can be used without registration, but if a user account is created, the system will save the entire image upload history. Although the Luigi database contains WSI information for each case, the query and the retrieved images handled by the system are patches extracted from WSIs because, in most situations, users submit small regions of a whole slide with specific features.

CBIR and prediction of genomic aberrations

When a query image is submitted to the Luigi system, 1,024-dimensional DTRs of the image, as well as the one image rotated 90° for imposing rotational invariance, are calculated. The system then searches for the most similar images in the database using the cosine similarity of the extracted DTRs between the query and database images. An approximate nearest neighbor search method with the randomized kd-tree algorithm implemented in pyflann (Muja and Lowe, 2009) was adopted to enable high-speed searching. For the prediction of genomic aberrations, the magnification level of the query image was first estimated by a Support Vector Machine classifier trained on the DTRs of cancerous regions for each cancer type in TCGA images at three magnification levels (128, 192, and $256 \mu\text{m}$). Subsequently the logistic regression classifier trained at the predicted magnification level was applied for each genomic aberration.

User interface of the luigi web system

Figure 7A shows the interface of the Luigi system. Users upload a histopathological image file by either dragging and dropping the file, pasting a screenshot of the image, or selecting “click here to select a file.” The uploaded image is resized to 256×256 pixels (the size of the images in the database). Because this resizing process could alter the texture, uploading images with appropriate magnification levels is essential to retrieve good results. Then, the Luigi system will retrieve the most relevant patch in each WSI from all cancer types, rank them according to cosine similarity and display them. Tissue origin can be changed using the pull-down menu above the results. When a user selects any of the retrieved images, that image will become a new query and can be used to retrieve relevant patches by clicking the “Search” button. The TCGA slide ID (TCGA-XX-XXXX-XXX-XX-XXX) accompanying each image contains a link to the cBioPortal website (Cerami et al., 2012), so users can open a new window showing the summary of the selected case. There users can review the pathology report as well as clinical and molecular information for the case.

Predicted genomic aberrations are shown in the right panel. The result from the estimated cancer type, i.e., the most frequent type found within the top 10 retrieved cases, is initially shown, but users can alter the cancer type manually.

Luigi is available at <http://luigi-pathology.com/>

Luigi smartphone application

A smartphone application was also developed to allow users to submit appropriately scaled query images directly to the Luigi web application and to view retrieved images. Figure 7C displays this interface. Users can take a photo of their cases or select an image stored in the smartphone, and then adjust the query image size while referring to the image on top. Users can then press the check button to display the relevant cases to the uploaded query image found for all tumors with the order of cosine similarity. The tissue of origin can be changed by selecting from the drop-down menu at the top of the screen. Users can also access to the cBioPortal website by tapping TCGA slide ID of the retrieved case. An iOS version of the smartphone application is currently available.

CBIR performance evaluation

We evaluated the performance of Luigi web systems to retrieve similar images in the following 12 cancer types: lower grade glioma (LGG), THCA, kidney renal clear-cell carcinoma (KIRC), lung adenocarcinoma (LUAD), breast cancer (BRCA), colon adenocarcinoma

(COAD), head and neck squamous cell carcinoma (HNSC), kidney chromophobe (KICH), thymoma (THYM), PRAD, sarcoma (SARC), and testicular germ cell tumor (TCGT). The query images representing typical morphology were selected from TCGA images by a trained pathologist, and retrieved images from all the cancer types were evaluated. We evaluated precision based on the cancer type except for LUAD, COAD, HNSC, in which adenocarcinoma or squamous cell carcinoma were used as the ground truth category. Precision values for the top 5, 10, and 20 images were evaluated. In another analysis, we prepared query images for 12 cancer types. We then created a dataset for each query containing 19 randomly selected images and the most relevant image selected by the Luigi. Five trained pathologists were asked to blindly select those most histologically similar to the query image, irrespective of cancer types.

Query images for the Luigi smartphone application were taken with a microscope mounted on an iPhone 8 using a Gosky Universal Cell Phone Adapter Mount (American Gosky Optics, GA, USA. <http://www.goskyoptics.com/>). We have selected 5 tumor regions from within our own WSIs—including human STAD and patient-derived tumor xenograft tissues from colorectal and pancreas adenocarcinoma—and used these as queries. The magnification levels were adjusted manually.

Estimation of the number of patients with predictable actionable mutations

We have roughly estimated the number of patients worldwide whose actionable mutations can be predicted solely via histology images. The numbers of incidences of each cancer type were downloaded from the Cancer Today website from the World Health Organization (<https://gco.iarc.fr/today/online-analysis-map>). The downloaded data contained crude incidence rates and were labeled with cancer types. Because some of the labeled cancer sites did not correspond to those of the TCGA's, additional processing was necessary. We have combined "lip, oral cavity," "hypopharynx," "larynx," "nasopharynx," and "oropharynx" cancer sites as being under the category of "HNSC." Because GBM and LGG in TCGA fall into "brain, central nervous system" for cancer sites, we estimated the incidence of each of these cancers using the average incidence rate of nervous system and intracranial tumors in England (Maile et al., 2016). For lung cancer, international incidence rates of lung adenocarcinoma and squamous cell carcinoma within all lung cancer types (Cheng et al., 2016) were used for estimation. Some of the cancer types in TCGA—such as TCGT, THYM, UCS, and SARC—that were not included in the WHO analysis, were not used due to a lack of statistics.

The latest available population data (mostly from 2018) in each country was downloaded from The World Bank (<https://data.worldbank.org/>).

We used actionable mutations with a positive predictive value (PPV) > 0.7 for the estimation. PPVs for each gene in specific cancer types were estimated from TCGA population as follows: (1) when PPVs based on an optimal classification threshold, which was determined by the Youden index (Youden, 1950) for the mean ROC curve among 3-fold cross-validations, exceed 0.7, the PPV and the corresponding true positive rate (TPR) was used. (2) Otherwise, when the maximum PPV in all the classification thresholds exceeded 0.7, the PPV and the corresponding TPR was used. The number of patients with predictable actionable mutations was estimated by the number of cancer incidence multiplied by the TPR of the mutation in TCGA population calculated for each of the corresponding PPVs. Table S8 shows the selected genes. The total number of patients who can be found to be actionable mutation-positive using our method was calculated by summing up all the estimated number of patients for each actionable gene or genomic aberration.

Notes on the risk of overfitting

Deep texture representations (DTRs) are produced by a deep neural network trained on general images. In the present study, the weights of the network were not optimized based on histology images. However, image patches from The Cancer Genome Atlas (TCGA) whole slide images (WSIs) were used to select the deep neural network model and its layers so that similarities based on the DTRs were correlated with those based on pathologists' judgments, as stated in the Methods and Figure S1A. Only 640 image patches were used for this selection and the queries' similarities with the top three images selected by each pathologist for each cancer type were used in calculations; however, this process does not preclude the possibility of overfitting through information leakage between the DTR selection and subsequent analysis. Therefore, we performed additional experiments to evaluate the possible effect of overfitting.

In Figure 2 [uniform manifold approximation and projection (UMAP) distribution of TCGA case, also shown in Figure S10A], we re-drew UMAP without the cases used for model and layer selection (hereafter, referred to as the "DTR training cases"). As shown in Figure S10B, the overall distribution remained almost unchanged. Although the relative position of thyroid cancer (THCA) cases moved, it continued to form a distinct cluster. Such movement of THCA cases was also observed in the UMAP distribution with the DTR training cases when a different random seed was used (Figure S10C); thus, the movement was likely due to the nature of the UMAP algorithm making use of randomness.

Considering Figure 3A [UMAP distribution and classification of prostate adenocarcinoma (PRAD) and stomach adenocarcinoma (STAD) cases], we retrained the classification model for PRAD and STAD classifications without the DTR training cases. We did not observe evident performance reduction; indeed, the area under the curve (AUC) increased from 0.91 to 0.93 in PRAD, whereas the AUC decreased from 0.94 to 0.92 in STAD.

Considering Figure 3B [UMAP distribution of pancreas adenocarcinoma (PAAD) cases], as shown in Figures S10A–S10C, the UMAP distribution for colorectal adenocarcinoma and PAAD was almost unchanged. We also confirmed that the DTR training cases did not contain the highlighted cases in Figure 3B.

Figure 5 (prediction of cancer-related genomic features), we retrained the classification model for annotated driver mutations with q-values <0.02 and without the DTR training cases; we then compared the AUC to the original result (Figure S10D). We did not

observe evidence of overfitting: the AUCs of the model with and without the training cases used in the model and layer selection were comparable, as shown in Figure S10D(i.e., only -0.02 lower on average in the model without the DTR training cases).

Considering Figure 6C (pathway mutations in THCA), we retrained the classification models without the DTR training cases. Following retraining, there was no evident performance reduction: AUC increased from 0.92 to 0.94 in RAS genes, it decreased from 0.91 to 0.90 in *BRAF*, and it decreased from 0.85 to 0.83 in RTK genes.

The removal of the DTR training cases could potentially cause negative effects on CBIR or classification in two ways: overfitting and reducing cases in the CBIR database or training cases for model development. However, we did not observe any apparent performance reduction in the additional experiments presented above. Moreover, DTRs functioned well for various external datasets, as shown in Figures 3C–3G and 5C. Taking these findings together, we conclude that the effect of overfitting was minimal, if it existed at all; thus, this study was not biased by overfitting.

QUANTIFICATION AND STATISTICAL ANALYSIS

Data were analyzed using R version 3.6.3. Statistical significance was determined using log rank test and multivariate Cox models with “survfit” and “survminer” libraries (Figures 3D and Table S3), Jonckheere-Terpstra trend test with “clinfun” library (Figure 3E), Fisher exact test with standard library (Figures 3F, S4A–S4B), and paired t test with standard library (“Performance comparison of driver mutation prediction with ShuffleNet-based classifier”). All statistical tests are two-sided and p value < 0.05 was considered statistically significant. In Figures 5A and S7, AUC scores from 100,000 random permutations were calculated and compared to the median AUC in 3-fold cross validation to calculate the p value. A “q-value” library was used to control for multiple testing and q -value < 0.02 was considered statistically significant.